Adventist University of Central Africa

AI-BASED CHATBOT FOR MENTAL HEALTH SUPPORT

Case study: Ndera neuropsychiatric teaching hospital

A final year project presented in partial fulfillment of the requirements for the degree of BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Major in Software Engineering

By

Mupenzi Serge

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ABSTRACT

Adventist University of Central Africa

A Final Year Project for the Bachelor's Degree of Science in Information Technology

Major Software Engineering

Title: AI-BASED CHATBOT FOR MENTAL HEALTH SUPPORT

Name of the student: Mupenzi Serge

Name of faculty Advisor: Mr. NSHUNGUYE Justin

Date Completed: November, 2025

Mental health is an increasingly important issue around the world, and Rwanda is no exception. Many people continue to face challenges in accessing timely and affordable care due to the shortage of professionals, high service costs, and the stigma that still surrounds mental illness. Although digital solutions have emerged, most lack meaningful, AI-driven interaction or an effective way to respond during a crisis. This project was created to help close that gap by developing an AI-based chatbot designed to offer accessible, confidential, and responsive mental health support anytime it's needed.

The main challenge was to design a system that could do more than just chat, it needed to truly understand and respond with empathy, while also recognizing when someone might be in emotional distress. The chatbot aims to provide personalized support through conversations, mood tracking, guided breathing exercises, and crisis response features.

Data was gathered through research studies, documentation, observation, and interviews with the patients to understand the real emotional. The Agile methodology was followed throughout development, allowing flexibility and improvement based on user experience. The system was designed with a user-centered approach and built using Next.js, TypeScript, and Tailwind CSS for the frontend, while Node.js and PostgreSQL power the backend for secure data handling and chatbot logic. This AI-powered chatbot serves as a caring digital companion that listens and responds with supportive guidance. Its purpose is to bridge the gap between therapy and self-help tools, making mental health support more accessible in Rwanda and beyond.

DECLARATION

I, **Mupenzi Serge** with Registration Number 25181, a student at the Adventist University of Central Africa, in the Faculty of Information Technology, Department of Software Engineering.

I hereby declare that this project, titled as "**AI-Based Chatbot for Mental Health Support**" is my own original work. I have carried out this project through personal research, development, and effort, and it has not been submitted for academic credit at AUCA or any other institution.

Signature: ……..

Date: …. /…./….

DEDICATION

I sincerely thank God for His grace and strength throughout this journey.

My gratitude goes to my family for their unwavering encouragement.

To my supervisor and IT faculty lecturers for their guidance.

I also appreciate mental health advocates and professionals

whose commitment continues

to inspire positive change.

APPROVAL

I, Mr. Nshunguye Justin, confirm that I have given my consent for this final year project report to be completed and submitted under my supervision.

Signature: …….

Date: …. /…./….

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LIST OF ABBREVIATIONS

AUCA:	Adventist University of Central Africa
RBC:	Rwanda Biometric Center
AI:	Artificial Intelligence
CBT:	Cognitive Behavioral Therapy
NLP/NLU:	Natural Language Processing/Natural Language Understanding
PTSD:	Post-Traumatic Stress Disorder
API:	Application Programming Interface
APA:	American Psychological Association
JWT:	JSON Web Token
REST:	Representational State Transfer
SQL:	Structured Query Language
ORM:	Object Relational Mapper
IDE:	Integrated Development Environment
SEO:	Search Engine Optimization
CSS:	Cascading Style Sheets
HTML:	HyperText Markup Language
PII:	Personal Identifiable Information
EHRs:	Electronic Health Records
GPT:	Generative Pre-Trained Transformer
PWA:	Progressive Web Application
UX/ UI:	User Experience/ User Interface
UML:	Unified Modeling Language
IT:	Information Technology
DB:	Database
ICT:	Information Communication Technology
BERT:	Bidirectional Encoder Representations from Transformers
WHO:	World Health Organization
RAM:	Random Access Memory
CHUK:	University Teaching Hospital of Kigali

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First and foremost, I thank God Almighty for His endless grace, wisdom, and strength that have guided me through this academic journey. Without His love and guidance, completing this project would not have been possible.

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I owe special gratitude to my supervisor, Mr. Justin Nshunguye, for his steady guidance, insightful feedback, and encouragement. Your mentorship not only helped me complete this work successfully but also inspired me to think critically and grow as a student and person.

I am profoundly thankful to my family, whose love, sacrifices, and support have been my greatest strength throughout my academic journey. Thank you for your patience during late nights, your wise guidance when I felt lost, and your belief in me even when I doubted myself. Your encouragement has been my motivation to keep going, and your values have shaped me.

To my siblings, thank you for your inspiration and continuous encouragement that pushed me to stay focused and achieve my goals.

I'm also deeply thankful to my collegues for their support, laughter, and motivation. Sharing this journey with you made it lighter and much more enjoyable.

Finally, to everyone who supported me in one way or another thank you. This achievement is as much yours as it is mine.

May God bless you abundantly!

Mupenzi Serge

CHAPTER 1

GENERAL INTRODUCTION

Introduction

Mental health has increasingly become one of the most urgent health challenges of our time, touching the lives of millions around the world, regardless of age, gender, or background. Research shows that one in four people will experience a mental health disorder at some point in their lives, with depression and anxiety being among the most common. In Rwanda, the challenge is equally significant about 1.2 million people are estimated to live with mental health issues. Sadly, many of them never receive the care they need due to financial limitations, the shortage of professionals, and the stigma that still surrounds mental illness.

With the rise of AI and natural language processing, new and creative solutions have emerged to bridge this gap. AI-powered chatbots have shown great potential in offering accessible, private, and round-the-clock mental health support. Unlike traditional therapy, these digital companions can provide immediate responses, maintain anonymity, and adapt to the emotional needs of each individual. They serve as a safe space for people who may feel hesitant to seek help in person.

This project, titled **"**AI-Based Chatbot for Mental Health Support," was created to address these challenges by developing a supportive and intelligent chatbot named SergeAI. The system aims to offer users personalized conversations, guided self-help exercises, mood tracking, and crisis response features. Beyond conversation, SergeAI is designed to listen, understand, and respond with empathy acting as both a guide and a companion for individuals navigating emotional distress.

The goal of this project is not just to build a technical solution but to make mental health care more approachable, inclusive, and stigma-free. By combining technology and compassion, SergeAI strives to reach people who may otherwise struggle in silence. This initiative also aligns with Rwanda's Vision 2050, which promotes digital transformation and improved access to healthcare for all. Through this project, technology becomes more than a tool it becomes a bridge to hope, healing, and human connection.

Background of the Study

In the aftermath of the 1994 Genocide against the Tutsi, Rwanda faced not only the enormous task of rebuilding its physical infrastructure but also the challenge of addressing deep psychological scars. Recognizing the importance of mental well-being for national recovery, the government placed mental health at the center of its post-genocide agenda. The adoption of the first national mental health policy around 1995 marked a pivotal step toward systematically addressing trauma and integrating mental health services into the broader healthcare system. This early commitment laid the foundation for a long-term effort to heal the population and promote mental wellness as a national priority. (health M. , 1995)

Over the ensuing decades, mental health gradually shifted from a marginal concern to a recognized pillar of health and national development. Policies such as the National Mental Health Policy (2020–2030) and the National Strategy for Health Professions Development 2030 reflect this evolution, establishing targets for workforce expansion like training 44 psychiatrists by 2030 and encouraging broader integration of mental health services into primary care. Despite these strides, the journey has been far from straightforward. A 2022 study showed that while approximately 61.7% of Rwandans knew where to seek mental health support, only around 5.3% had accessed formal services. Barriers such as stigma, limited awareness, and shortages of trained professionals continue to hinder access, particularly in rural and remote areas. The lingering effects of trauma, coupled with societal stressors including economic pressures and pandemics, continue to shape the context in which mental health services operate. (RBC, 2022)

The Ndera Neuropsychiatric Teaching Hospital stands as Rwanda's flagship institution for mental health care. As the country's primary psychiatric referral center and a key teaching hospital, it carries the dual responsibility of providing specialized care to patients while nurturing the next generation of mental health professionals. However, the hospital operates under significant constraints. National data indicate that Rwanda has only about 16 psychiatrists in the public sector, roughly one psychiatrist per 100,000 people. This shortage places immense pressure on Ndera to manage high patient loads, provide a wide spectrum of services from trauma care to substance use treatment, and simultaneously fulfill its educational and research mandates.

In this context, digital, scalable solutions such as an AI-based chatbot for mental health support are particularly relevant. Such a tool can extend the reach of Ndera's services beyond the hospital's walls, providing continuous support when the core workforce is stretched, and complementing the hospital's mission of inclusive, accessible care. By connecting into Ndera's ecosystem, whether as a referral adjunct or community support layer, the chatbot can help reduce the burden on the institution, improve early access, and deliver culturally relevant support aligned with Rwanda's national mental health system.

This project responds directly to the need for innovation in Rwanda's mental health system by developing an AI-based chatbot tailored for mental health support. By offering 24/7 assistance, mood tracking, and conversational interaction, the chatbot helps bridge temporal and geographic gaps in service provision, which is particularly important for individuals outside urban areas or when professionals at institutions like Ndera are unavailable.

Leveraging modern web technologies and AI, including natural language processing, the system responds dynamically to user inputs and adapts to Rwanda's cultural, linguistic, and societal context, addressing the mismatch often observed in global tools designed for Western settings that fail to resonate locally.

The solution is designed to complement existing institutional frameworks rather than replace them, supporting referral flows, generating data for oversight, and allowing longer-term scalability. By introducing a digital layer that works alongside human professionals, it helps strengthen Rwanda's mental health infrastructure.

Ultimately, the project aims to contribute to Rwanda's goal of inclusive mental health care by offering a scalable, culturally sensitive, and technology-enabled tool. It seeks to reduce stigma, promote early help-seeking behavior, and relieve pressure on core institutions like Ndera, advancing the nation's capacity to support mental well-being for all.

Statement of the problem

The aim of this study is to design and develop an AI-based Mental Health Support Chatbot that will modernize Rwanda's approach to mental health care by addressing gaps in accessibility, affordability, and timely support. The system will provide a digital, user-friendly platform offering continuous guidance, mood and symptom tracking, confidential communication, and intelligent features such as culturally sensitive responses and early crisis alerts. By bridging the gap between need and access, this platform will directly address systemic and societal barriers, improving accessibility, timeliness, privacy, and overall effectiveness of care.

Key mental health challenges facing Ndera teaching hospital and Rwanda's mental health system:

Rwanda's mental health system, with Ndera Neuropsychiatric Teaching Hospital at its core, continues to face significant structural, social, and technological challenges that limit effective service delivery and equitable access to care.

One of the most pressing issues is the severe shortage of trained mental health professionals: With fewer than 60 psychiatrists serving a population of over 13 million people most of whom are concentrated in urban centers Ndera Hospital remains under immense pressure as the national referral facility. This shortage leaves rural and remote communities largely underserved, forcing patients to travel long distances at high personal and financial cost or rely on traditional practices that may not adequately address conditions such as depression, trauma, and anxiety.

Limited accessibility and continuity of care further exacerbate the problem: Many individuals are unable to access ongoing, personalized mental health support due to distance, long waiting times, and the unavailability of specialists. These barriers are intensified by high costs, as private consultations and therapy sessions are often unaffordable, while indirect and hidden costs within public facilities still pose financial challenges for many families.

Stigma and privacy: concerns remain deeply rooted societal issues that significantly affect help-seeking behavior. Mental illness is frequently perceived as a sign of weakness or spiritual failure,

discouraging individuals particularly young people and professionals from seeking early support. Fear of social judgment and lack of confidential avenues for care contribute to delayed treatment and worsening mental health outcomes.

Another critical challenge is inadequate crisis intervention capacity: Few 24/7 services exist to respond promptly to emergencies such as suicidal ideation, severe depression, or acute psychological distress. As a result, caregivers and institutions like Ndera Hospital often struggle to identify and respond to high-risk cases in a timely manner, placing vulnerable individuals at serious risk.

The effectiveness of many existing digital mental health tools is also limited by a lack of cultural and contextual relevance: Most global solutions do not adequately reflect Rwanda's linguistic, cultural, and social realities, reducing their usefulness for local users and limiting adoption. This gap underscores the need for culturally adapted, context-aware mental health support systems.

Finally, there is limited coordination between digital solutions and existing mental health institutions: The absence of integrated digital platforms that complement facilities like Ndera Hospital leads to fragmented care, inefficient referrals, and increased institutional burden. Without proper evaluation mechanisms, it is also difficult to measure the impact of emerging digital tools on access, crisis response, stigma reduction, and alignment with national mental health goals.

The chatbot addresses these challenges by providing continuous, confidential, and culturally relevant support, along with real-time alerts and early detection of distress. By centralizing mental health support digitally, the system improves accessibility, reduces barriers, and strengthens Rwanda's overall mental health capacity. It empowers individuals, families, and communities to engage in proactive care, enhances the efficiency of existing services, and aligns with national goals for inclusive, sustainable mental health care. Ultimately, the platform aims to create a resilient, technology-enabled mental health ecosystem accessible to every Rwandan, promoting well-being and social stability across the country.

Choice and Motivation of the Study

This project was inspired by the urgent need for accessible, personalized, and stigma-free mental health support in Rwanda. Despite growing awareness, many people especially in rural areas still face barriers such as high costs, limited professionals, and social stigma when seeking help. The idea of developing an AI-based chatbot emerged as a practical solution to bridge this gap by offering 24/7 emotional support, personalized guidance, and self-help resources to anyone, anywhere, at any time.

To AUCA: This project reflects the mission and values of the AUCA, which emphasizes the use of technology to serve humanity with compassion and excellence. It allowed me to apply the technical and ethical knowledge gained throughout my studies in Software Engineering to solve a real social problem. By integrating AI**,** data management, and user-centered design, this chatbot demonstrates how AUCA students can contribute to digital transformation in healthcare, making mental health services more inclusive, efficient, and empathetic.

To Ndera Neuropsychiatric Teaching Hospital: The Ndera Neuropsychiatric Teaching Hospital, Rwanda's main mental health facility, faces growing demand for services with limited professional capacity. The chatbot acts as a complementary virtual assistant, helping users manage stress, anxiety, and depression while connecting them to professional help when necessary. It also includes crisis response and mood-tracking features, which can ease the hospital's workload and support community-based care initiatives through technology-driven engagement.

To the Student: For me, as a final-year Software Engineering student, this project has been both a technical challenge and a personal calling, reflecting my belief that technology should also heal and empower people. It strengthened my skills in AI, NLP, and ethical design, and taught me the importance of creating systems that show empathy. This work supports Rwanda's digital vision and represents my effort to inspire future AUCA innovators to build solutions that make a meaningful impact.

Objectives of the Study

General Objective

The overall goal of this project is to design and implement an AI-powered chatbot that delivers accessible, personalized, and effective mental health support for individuals in Rwanda. By leveraging innovative technology, the system seeks to address the gaps in the current mental health service landscape and provide scalable, user-friendly solutions, develop a robust AI-based chatbot system that provides real-time, personalized mental health support through conversational interaction, mood tracking, and self-help tools.

Specific Objectives

- Develop a secure, AI-powered mental health support platform that allows users to access personalized guidance, emotional support, and self-assessment tools through web and mobile interfaces, ensuring accessibility and confidentiality for all users.
- Design a user-friendly interface that ensures easy navigation and accessibility for users of all backgrounds, including those in rural communities, while maintaining an engaging and empathetic experience.
- Implement a robust and secure backend and database system to store user interactions, mood assessments, and progress history with strict data privacy, encryption, and compliance with ethical standards, ensuring scalability and reliability for multiple simultaneous users.
- Offer detailed and reporting dashboards to help mental health practitioners and administrators monitor engagement trends, track user progress, and make informed, data-driven decisions for improving service delivery.
- Implement crisis intervention protocols that detect users in distress and automatically link them to professional mental health services or emergency contacts for timely assistance.
- Enhance early detection and prevention capabilities by incorporating predictive analytics and risk indicators that help identify emerging mental health concerns before they escalate into severe crises.

Scope of the Project

This study is limited by several system technical, and operational constraints that shape the functionality and applicability of the AI-Based Mental Health Support Chatbot. While the system offers digital emotional support, guided conversations, and mood-tracking features, its capabilities remain confined to non-clinical use. The chatbot provides general wellbeing assistance but does not replace licensed mental health professionals or serve as a diagnostic tool. The system is designed for adults with mild to moderate mental health concerns, excluding individuals requiring intensive psychiatric care, crisis intervention, or medically supervised treatment.

The functionality of the chatbot also depends heavily on internet connectivity, as real-time AI processing and response generation require access to cloud-based services. Limited or unstable network access may reduce the system's effectiveness. Additionally, the chatbot currently supports only text-based interactions, as features such as voice recognition, video sessions, or multimedia-based therapy modules are not included in the initial release. The system also does not integrate with electronic health record (EHR) systems, medical databases, or external clinical platforms, which restricts its use in formal healthcare settings and collaborative care environments.

In terms of AI performance, the accuracy of sentiment analysis and emotional interpretation may be reduced due to the lack of locally trained datasets that reflect Rwandan linguistic, cultural, and contextual nuances. This limitation may affect the system's ability to fully understand user expressions, idioms, or culturally influenced emotional cues. Furthermore, the chatbot does not provide legal advice, prescribe medication, or issue emergency interventions beyond displaying referral information and emergency contacts.

Overall, these limitations highlight the boundaries of the system as a supportive digital tool rather than a clinical mental health solution. Future system improvements may address these gaps by expanding functionality, enhancing AI training data, and exploring integration with professional mental health services.

Methodology and Techniques Used in the Study

This project uses a mixed-methods approach, combining both qualitative and quantitative research techniques with modern software development practices. This ensures a well-rounded process for designing, developing, and validating the system.

Documentation

Documentation is a qualitative research method that involves systematically collecting, reviewing, and analyzing existing written materials such as reports, records, policy documents, and prior studies related to a research topic. It enables researchers to understand the background of the problem, identify gaps and existing challenges, and obtain reliable secondary data that informs and guides the development of practical and evidence-based solutions (Creswell, 2018)

In this study, I used documentation to review reports and publications from the RBC and the Ministry of Health. These sources provided important insights into Rwanda's mental health policies, service delivery challenges, and the current gaps in accessibility and awareness. This helped guide the development of the AI-Based Mental Health Support System to ensure it aligns with national priorities and supports ongoing efforts to improve mental health care in Rwanda (RBC, 2022)

Interview

Interviews are a qualitative research method involving structured or semi-structured conversations with participants to collect in-depth insights about their experiences, perceptions, and challenges. This method allows researchers to understand issues from the viewpoints of those directly affected, offering detailed and personal perspectives that may not be revealed through documentation or observation alone.

In this study, interviews were conducted with both mental health professionals and potential users of the system to gather practical insights into the accessibility and quality of mental health services in Rwanda. Mental health professionals highlighted the shortage of trained specialists, limited digital tools for remote support, and the overwhelming patient load in facilities like Ndera

Neuropsychiatric Teaching Hospital. Meanwhile, potential users especially students and young adults expressed the need for a private, confidential, and easily accessible platform to seek mental health support without fear of stigma or judgment.

I, Mupenzi Serge, a finalist in AUCA, IT Faculty**,** Department of Software Engineering, am conducting this project on the development of an AI-Based Mental Health Support System for Rwanda. The project seeks to improve accessibility, privacy, and continuity of mental health care through digital innovation and AI-assisted support.

Below are the key questions asked during the interviews, along with summarized responses:

Question 1: What are the main challenges people meets when visiting the mental health services? **Answer:** Many participants mentioned limited access to qualified professionals, long distances to reach hospitals, and high service costs. Rural communities are particularly affected, with few mental health centers and a shortage of trained counselors.

Question 2: How does stigma affect people's willingness to seek mental health support? **Answer:** Respondents explained that stigma is a major barrier. People often fear being labeled or judged, which discourages them from seeking therapy or counseling. This is especially true for young adults and professionals who worry about their reputation.

Question 3: What role do you think technology could play in improving mental health support? **Answer:** Participants felt that technology could make mental health support more accessible and private. A digital platform can provide a safe space for users to express themselves, learn coping strategies, and get guidance without fear of being seen or judged.

Question 4: What features should an AI-based mental health support system include? **Answer:** Suggested features included an anonymous chat interface, mood tracking tools and referral options to professionals for serious cases.

Question 5: What concerns do you have about using AI for mental health care? **Answer:** Participants expressed concerns about data privacy, reliability of automated responses,

and the limited human interaction. They emphasized that confidentiality is essential and that the system should complement, not replace, human mental health professionals.

Question 6: How can this project contribute to improving mental health care in Rwanda? **Answer:** Interviewees agreed that the project could bridge the gap between awareness and access by offering 24/7 support, reducing stigma through anonymity, and encouraging early help-seeking behavior. They also noted that it could relieve pressure on professionals and complement services at facilities like Ndera.

Question 7: Can you tell me how comfortable you feel talking about your mental health with others?

Answer: Honestly, it's not always easy. I often worry that people won't understand or that they'll judge me. In my community, when you talk about mental health, some think you're just overreacting or that you're weak. That makes it harder to open up. I'd probably feel more comfortable if there were private and understanding spaces like online chats or anonymous support platforms where I could talk freely without being identified.

Expected Results

Expected results refer to the anticipated benefits and measurable improvements that the implementation of the AI-Based Chatbot for Mental Health Support will bring to users, mental health service providers, and the broader digital health ecosystem. These outcomes reflect enhancements in accessibility, emotional wellbeing support, user engagement, crisis response, and overall mental health service delivery.

Upon successful implementation of the system, the following outcomes are expected:

Improved Access to Mental Health Support: The chatbot will provide 24/7 availability, ensuring that users can receive emotional support at any time without waiting for appointments or service hours.

Reduced Stigma Through Anonymous Support: Users will be able to engage with the chatbot anonymously, lowering psychological barriers and encouraging individuals who might avoid traditional therapy to seek help.

Enhanced User Self-Awareness: Through mood tracking features and visual emotional analytics, users will gain better insight into their emotional patterns, triggers, and long-term wellbeing trends.

Immediate Emotional Assistance: Guided self-help tools such as breathing exercises will offer instant support to users experiencing stress, anxiety, or emotional discomfort.

Early Crisis Detection: Built-in sentiment analysis and crisis detection mechanisms will help identify signs of distress early, enabling timely referrals to emergency contacts or professional services.

Improved User Engagement: Personalized conversation flows based on user history and emotional trends will increase relevance and encourage consistent interaction with the system.

Secure and Centralized Data Management: User information, mood logs, and conversation histories will be stored securely, providing organized and confidential records for future reference and system improvement.

Reduced Pressure on Mental Health Professionals: By handling general wellness inquiries and low-risk emotional concerns, the chatbot will help reduce the workload on mental health service providers, allowing professionals to focus on more severe cases.

Data-Driven Mental Health Insights: Aggregated user data will allow identification of common emotional trends, stress patterns, and frequently reported challenges, supporting evidence-based improvements in digital mental health services.

Support for Long-Term Wellbeing: Through continuous access to emotional guidance and self-management tools, users are expected to develop healthier coping strategies, improved resilience, and increased awareness of when to seek professional help.

Increased User Satisfaction: The combination of instant support, personalized conversations, and non-judgmental interaction is expected to enhance user satisfaction and encourage long-term use of the platform.

Better Monitoring of Emotional Progress: Users will be able to track improvements or declines in their wellness over time, enabling better personal reflection and timely decision-making about seeking professional help.

Greater Inclusion of Underserved Populations: Because the platform is low-cost, mobile-friendly, and accessible online, individuals in rural or underserved areas will gain access to mental health support they may not have otherwise received.

Improved Digital Health Literacy: Regular interaction with digital mental health tools will increase users' familiarity with technology-based support systems, contributing to broader digital health adoption in Rwanda.

Organization of the work

Chapter One: Introduces the study by providing background on mental health challenges, the problem statement, motivation, objectives, scope, methodology, and anticipated outcomes.

Chapter Two: Examines the current mental health support landscape in Rwanda, identifies service gaps, and reviews literature on AI-based interventions to guide the system's development.

Chapter Three: Focuses on system planning and design, detailing functional and non-functional requirements, system architecture, AI integration, and supporting diagrams.

Chapter Four: Covers the development and testing of the SergeAI platform, including the technologies used, coding approaches, AI implementation, and challenges faced.

Chapter Five: Summarizes the key findings, reflects on the project's contributions to mental health support, and highlights recommendations for future enhancements and scalability.

CHAPTER 2

ANALYSIS OF THE EXISTING SYSTEM

Introduction

This chapter examines the current state of mental health support systems in Rwanda, exploring both traditional and digital approaches to identify gaps and opportunities for technological improvement. The goal is to understand existing challenges and show how an AI-based chatbot could strengthen access to mental health care and provide emotional support for those in need.

Over the years, Rwanda has made significant progress in promoting mental health through national initiatives led by institutions such as the Ndera Neuropsychiatric Teaching Hospital and the Rwanda Biomedical Centre. Despite these efforts, many individuals still face barriers including limited access to specialists, high costs, and the persistent stigma surrounding mental health conditions. These challenges are particularly acute in rural areas, where professional help is often scarce and difficult to reach.

Globally, digital health innovations have demonstrated great potential in addressing similar challenges. AI-powered chatbots, online counseling platforms, and self-help mobile applications are increasingly used to provide continuous, affordable, and confidential support. However, in Rwanda, adoption of such technologies remains minimal, leaving a gap between the growing mental health needs and the solutions currently available.

This analysis highlights key shortcomings in the current system, such as inadequate follow-up care, limited service coverage, and weak crisis intervention mechanisms, while also evaluating national efforts to improve awareness and accessibility.

Drawing from government reports, research studies, and stakeholder insights, this chapter lays a strong foundation for the proposed AI-based chatbot. It emphasizes the importance of using technology to complement existing services, ensuring that mental health care becomes more inclusive, accessible, and responsive to the needs of all Rwandans.

Description of the Existing System Environment

Historical Background

Rwanda's mental health sector has undergone significant development since the mid-1990s. Following the 1994 Genocide against the Tutsi, the government prioritized mental health as a key component of national recovery and social cohesion. Early initiatives focused on integrating mental health services into general healthcare facilities and establishing specialized institutions, with the Ndera Neuropsychiatric Teaching Hospital serving as the central hub for psychiatric care in the country.

Over time, the Rwandan government introduced several policies and frameworks to strengthen mental health services. The National Mental Health Policy (2020–2030) highlights two main priorities: expanding community-based mental health programs and integrating mental health care into primary health services. Despite these efforts, the sector continues to face challenges such as limited professional capacity, uneven service distribution, and persistent societal stigma that discourages many individuals from seeking help.

Digital health solutions have started gaining momentum in Rwanda, supported by the country's push to become a knowledge-based economy. With over 75% of the population now accessing mobile internet, opportunities for mobile and AI-driven mental health tools are expanding rapidly. However, most existing solutions are generic or designed for Western populations, limiting their effectiveness in Rwanda's cultural, linguistic, and socioeconomic context.

The increasing adoption of AI and NLP presents new possibilities for enhancing mental health support. Globally, AI-based chatbots have demonstrated success in providing accessible, anonymous, and personalized assistance, including guidance, emotional support, and mood tracking. Implementing these technologies in Rwanda offers a chance to complement traditional services, reduce the burden on limited mental health professionals, and improve access for underserved communities.

Mission

To provide accessible, culturally relevant, and stigma-free mental health support to all Rwandans through innovative digital solutions that complement existing healthcare services.

Vision

To become a leading platform for mental health care in Rwanda, offering continuous, empathetic, and evidence-based support, while contributing to a stronger, more resilient, and mentally healthy society.

Description of the Existing System

The current mental health support system in Rwanda operates primarily through hospital-based services, community programs, and limited digital initiatives. Most psychological and psychiatric support is delivered in person at health centers, district hospitals, or specialized institutions such as the Ndera Neuropsychiatric Teaching Hospital. Clients are required to visit these facilities physically for consultations, therapy sessions, and medication refills. Follow-ups are typically conducted through in-person appointments or basic phone communication, which makes continuous support difficult for those living far from service centers.

While community-based programs exist, they rely heavily on outreach campaigns, group discussions, and NGO-led initiatives to promote awareness and reduce stigma. These programs often lack structured digital systems for data collection, referral tracking, or individualized follow-up. As a result, people experiencing mental distress frequently go unmonitored after initial counseling or medication.

Key Issues with the Existing System:

Limited Accessibility: Access to mental health professionals is largely concentrated in urban areas, forcing individuals in rural regions to travel long distances to receive care. This uneven distribution limits early diagnosis and continuous therapy, particularly for low-income populations who cannot afford frequent travel.

High Cost of Services: Despite public insurance coverage, the overall cost of mental health care including private therapy sessions, medication, and transportation remains high for many. This financial burden discourages consistent treatment follow-up and contributes to a large number of untreated mental health conditions.

Lack of Continuous Support: After initial counseling or hospital discharge, most patients receive minimal follow-up. There are no digital systems in place for tracking progress, emotional check-ins, or ongoing support, increasing the likelihood of relapse and reducing long-term recovery success.

Inadequate Crisis Intervention: Mechanisms for immediate response during mental health emergencies such as suicidal thoughts or panic attacks are limited. The absence of 24/7 helplines or automated crisis support systems leaves individuals vulnerable and without help during critical moments.

Stigma and Privacy Concerns: Social stigma remains a major barrier that prevents people from seeking professional help. Additionally, the lack of anonymous and confidential digital platforms discourages users from openly sharing their struggles, perpetuating silence and untreated distress.

Limited Digital Integration: Existing digital platforms do not effectively integrate with national health systems or hospital databases, resulting in data fragmentation. This lack of coordination makes it difficult for policymakers and health professionals to monitor national mental health trends, evaluate program effectiveness, and allocate resources efficiently.

Lack of Real-Time Support and Reporting: Current systems lack real-time feedback, monitoring, and data reporting mechanisms. As a result, mental health professionals face challenges in tracking patient progress, identifying urgent cases, and providing timely interventions.

These challenges collectively reduce the efficiency, inclusivity, and responsiveness of Rwanda's mental health care system. Without digital innovation and AI-driven solutions, individuals will continue to face long wait times, limited access, and inadequate crisis response, highlighting the urgent need for an intelligent mental health support platform that bridges these critical gaps. Such a system would not only extend support to underserved communities but also ease the growing pressure on mental health professionals. By providing timely, user-centered assistance, it can strengthen the overall continuity and quality of mental health care in the country.

Analysis of Existing System

The current mental health support system in Rwanda relies heavily on traditional, in-person care at hospitals and health centers. While these facilities play a crucial role in providing treatment, the overall system faces several challenges related to accessibility, communication, and continuity of care.

Dependence on In-Person Care: The system primarily depends on physical visits to hospitals and health centers for diagnosis, therapy, and medication. This approach limits access for individuals who cannot easily travel due to distance, cost, or stigma, often delaying early intervention and support.

Limited Communication Methods: Communication between patients and healthcare providers is mainly face-to-face or through occasional phone calls. There are no structured digital communication channels for counseling, follow-ups, or progress tracking, which reduces the system's ability to maintain consistent support.

Lack of Continuous Follow-Up: After receiving initial treatment or counseling, many patients do not benefit from regular follow-ups. This lack of ongoing monitoring increases the likelihood of relapse or discontinuation of therapy, as patients are often left to manage recovery alone.

Community Programs Are Basic: While community-based programs exist, they mainly focus on raising awareness and offering general emotional support. Few provide structured or personalized care plans, and most community workers lack the professional training needed to handle complex mental health issues.

Accessibility Challenges: Due to the concentration of mental health professionals in urban areas, individuals from remote regions must travel long distances for consultations or therapy. These accessibility barriers, combined with financial constraints, make it difficult for many to seek timely and consistent mental health care.

Modeling the current system

The diagram illustrates the mental health care workflow for patients experiencing psychological distress, showing interactions across different levels of care from local health centers to district and Ndera hospitals. It captures the decision-making process based on the severity of the patient's condition, including counseling, medication, group sessions, or advanced therapy. Patients with moderate conditions receive support at local health centers, while tougher cases are escalated to higher-level facilities for advanced screening and specialized care. Follow-up procedures ensure continuity of care, and each stage emphasizes structured referrals, monitoring, and patient engagement to improve mental health outcomes

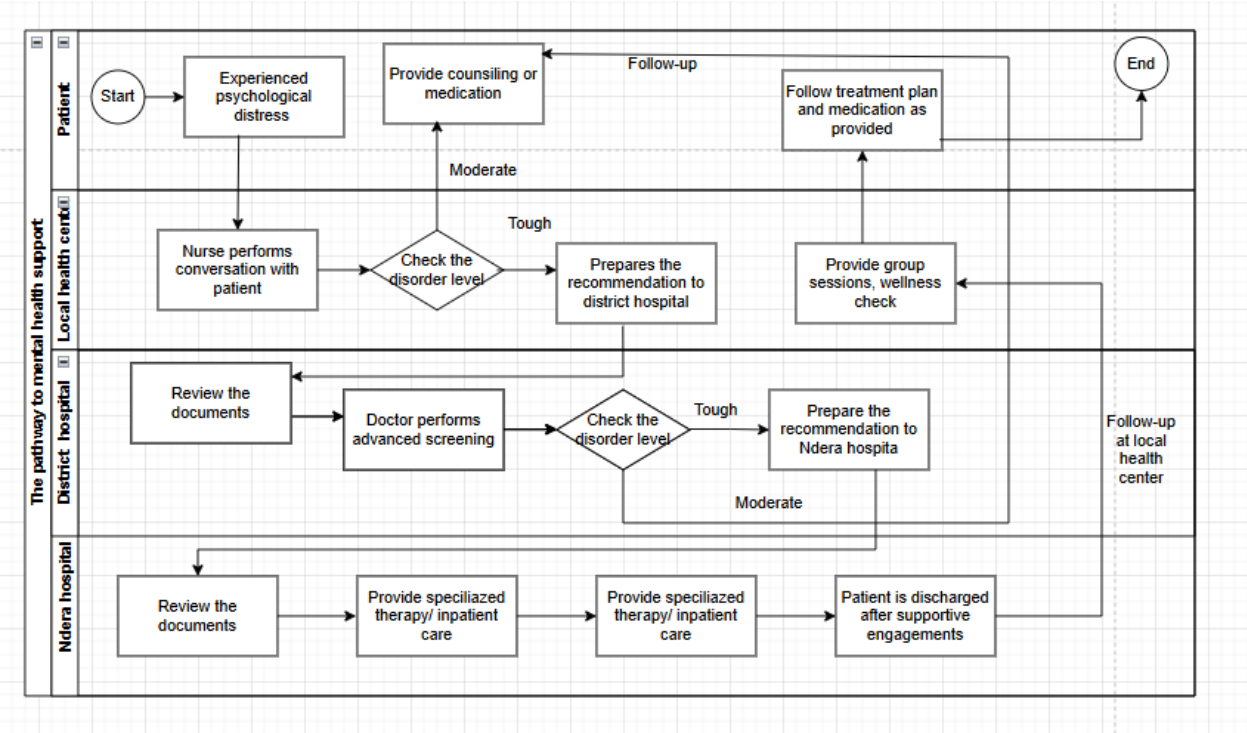

Figure 1: Current system

Here is an overview of the Current mental health support

Step 1: A Person Reaches Out for Help**:** The journey begins when an individual, or their concerned family, recognizes the need for support. They take the brave first step of visiting their local health center, often encouraged by a trusted Community Health Worker from their village.

Step 2: Initial Assessment by a Nurse: At the local health center, a compassionate nurse sits with the person to understand their struggles. This initial screening is a conversation to gauge the severity of their emotional or psychological distress.

Step 3: Care at the Community Level: If the condition is mild, the nurse provides immediate comfort and care. This may include basic counseling, reassurance, or a short course of medication to help manage the symptoms, allowing the person to recover within their community.

Step 4: Referral for Specialized Care: For more complex or severe cases that need advanced attention, the nurse prepares a referral. This is a formal recommendation for the person to seek specialized treatment at a higher-level facility, starting with the district hospital.

Step 5: Evaluation by a General Practitioner: At the district hospital, a doctor conducts a deeper assessment. They have more training and resources to manage a wider range of mental health conditions, providing a more detailed diagnosis and treatment plan.

Step 6: Escalation to National Specialists: If the condition is highly complex, persistent, or requires hospital admission, the doctor refers the person to Ndera Neuropsychiatric Teaching Hospital. Here, they will receive care from the country's top mental health specialists and psychiatrists.

Step 7: Specialized Treatment and Recovery: At Ndera Hospital, the individual receives a comprehensive treatment plan tailored to their needs. This can include specialized therapy, medication management, and inpatient care, all focused on helping them achieve stability and recovery.

Step 8: Return to Community with a Plan: Once their health is stabilized, the person is discharged back to their home community. They return with a clear follow-up plan to ensure their recovery continues.

Step 9: Community Follow-Up and Support: Back in their community, local health workers and support programs step in to provide ongoing group sessions, reintegration support, and wellness checks. This final step is crucial for long-term healing and preventing relapse, ensuring no one is left behind on their road to recovery.

Problems of the Existing System

Performance

- The process involves multiple in-person steps, causing delays and limited throughput.
- Lack of real-time communication and follow-up support prolongs treatment and recovery times.
- Manual record-keeping increases the risk of lost or incomplete information, reducing overall-efficiency.

Information

- **Input**: Patients often have to provide repeated or incomplete information at each visit.
- **Output**: Updates on treatment progress are communicated manually, limiting transparency.
- **Storage**: Fragmented paper-based or basic digital storage makes data management cumbersome.

Economics

- The system requires high operational costs due to repeated in-person visits.
- Travel costs and lost work hours for patients increase the financial burden, particularly for rural-populations.

Control

- Limited mechanisms exist for monitoring patient progress between visits.
- Crisis detection and timely intervention are insufficient, reducing overall system effectiveness.

Efficiency

- Coordination between patients, healthcare providers, and community programs is largely manual.
- Absence of integrated digital tools prevents real-time tracking and workflow optimization.

Service

- Patients receive support only during scheduled sessions, with no immediate help available during off-hours.
- Communication channels are inconsistent, relying on in-person visits, phone calls.
- Lack of transparency and real-time feedback reduces patient satisfaction and trust.

Proposed Solutions

Building on the analysis of current system limitations, the AI-Based Chatbot for Mental Health Support offers a comprehensive technological approach designed to address key gaps while complementing existing mental health services.

Accessibility enhancement: focuses on providing 24/7 availability of mental health support, the AI chatbot ensures that users can receive immediate help anytime and anywhere without needing appointments or waiting for available professionals.

Cost-effectiveness: offering free access to basic mental health care and early intervention, it removes financial barriers that often prevent people from seeking help. The chatbot's scalability allows many users to access services simultaneously without additional operational costs, ensuring sustainability in the long term.

Community integration: strengthens the system by connecting digital tools with existing local resources. The chatbot complements, rather than replaces, community health workers by providing referral pathways for those needing in-person follow-up.

System Requirements

Based on the identified gaps and proposed solutions, the system requirements for the AI-Based Chatbot for Mental Health Support are outlined below:

Functional Requirements

The functional requirements outline what the AI-Based Mental Health Chatbot System must do to provide effective, secure, and compassionate digital mental health support. These requirements

focus on communication, user management, and integration with emergency and community support systems.

- **REQ 1:** The system shall process and respond to user inputs in English, ensuring natural and contextually relevant conversations.
- **REQ 2:** The chatbot shall maintain conversational context across multiple exchanges to create coherent and continuous dialogue with the user.
- **REQ 3:** The system shall recognize and respond appropriately to emotional expressions, stress indicators, and common mental health terminology.
- **REQ 4:** The chatbot shall provide empathetic, supportive, and non-judgmental responses aligned with therapeutic communication principles.
- **REQ 5:** The system shall offer accurate information about common mental health conditions, including depression, anxiety, stress, and trauma.
- **REQ 6:** The chatbot shall guide users through evidence-based self-help techniques such as deep breathing, mindfulness, and relaxation tips.
- **REQ 7:** The system shall detect signs of suicidal ideation, self-harm, or severe emotional distress through language pattern analysis.
- **REQ 8:** The system shall provide immediate access to crisis resources, including hotlines and emergency contact information relevant to the user's location.

User Management and Personalization

- **REQ 9:** Users shall be able to create personal accounts using email or username and secure password authentication to protect their data and sessions.
- **REQ 10:** The system shall manage user sessions securely, ensuring data confidentiality and session timeout after periods of inactivity.
- **REQ 11:** The chatbot shall allow users to record and track their daily mood using visual scales or short reflective prompts.
- **REQ 12:** The system shall generate personalized mood trend reports and insights, helping users monitor their emotional well-being over time.

- **REQ 13:** The system shall integrate with local emergency services, such as medical teams, police, or mental health hotlines, for immediate support.

Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards that ensure the system is reliable, accessible, and user-friendly. These characteristics support operational efficiency, user trust, and long-term sustainability of the chatbot.

Performance

- **REQ 14:** The system shall maintain a maximum response time of under 2 seconds during standard user interactions.
- **REQ 15:** The chatbot shall ensure a minimum uptime to guarantee consistent availability for users seeking support.

Scalability

- **REQ 16:** The platform architecture should be scalable to accommodate increasing numbers of users

Usability and Accessibility

- **REQ 17:** The chatbot interface shall be intuitive and require no prior technical training for users.

Security

- **REQ 18:** The chatbot shall comply with data protection regulations and ethical guidelines for handling sensitive mental health information.

CHAPTER 3

REQUIREMENTS ANALYSIS AND DESIGN OF THE NEW SYSTEM

Introduction

The success of any information system depends on how well it meets user needs and addresses shortcomings in existing solutions. For this AI-Based Mental Health Support System, the processes of requirement analysis and system design are essential to ensure the platform is functional, secure, and user-friendly for individuals seeking mental health assistance.

Requirement analysis identifies what the new system must achieve by gathering and documenting the needs of users and administrators. This includes functional requirements, such as personalized support, self-assessment tools, and crisis response, as well as non-functional requirements like usability, performance, security, and accessibility. By examining gaps in current solutions, this analysis ensures that the system can provide timely, inclusive, and culturally relevant mental health support.

System design translates these requirements into structured models and frameworks that guide development. It defines the architecture of the platform, including the user interface, database, and AI components, ensuring that all elements work together efficiently and can adapt to future needs. Design considerations also prioritize clarity, ease of navigation, and security to enhance user experience.

Through thorough requirement analysis and thoughtful system design, this project aims to deliver a solution that is both technologically robust and socially relevant. The AI-Based Mental Health Support System will provide accessible, personalized, and secure assistance, contributing to improved mental health support and responsiveness in Rwanda.

Design Overview

The design phase moves from understanding the current shortcomings of mental health service delivery to creating a digital, user-friendly solution that bridges accessibility and emotional care. It converts both functional and non-functional requirements into a clear system architecture and interface plan that ensures inclusivity, responsiveness, and data security.

A well-designed AI-Based Mental Health Support System ensures that users can privately interact with the chatbot, receive immediate assistance, and access local mental health resources without barriers. The system emphasizes empathy, cultural sensitivity, and continuous support while maintaining technical reliability, scalability, and security.

The design process focuses on two major components:

System Analysis: Examines the limitations of existing hospital-based mental health support systems and the absence of continuous digital engagement for users, especially those in rural areas.
System Design: Defines the architecture, interfaces, and modules that deliver seamless interaction between users and the AI system.

The AI-Based Mental Health Support System follows best practices in software engineering and applies object-oriented design principles supported by UML modeling techniques. The system is designed to be modular, secure, scalable, and adaptable, ensuring that it can evolve alongside the growing mental health support needs in Rwanda. By integrating modern design methodologies with user-centered principles, AI-Based Mental Health Support System provides a reliable digital framework that promotes accessibility, confidentiality, and emotional well-being for all users.

Design of the New System

This section presents the detailed design of the AI-Based Mental Health Support Chatbot, transforming analyzed requirements into a practical system architecture and clearly defined functional components. The system is designed to deliver accessible, responsive, and culturally sensitive mental health support through conversational AI, helping users manage stress, anxiety, and emotional well-being in real time.

The proposed system integrates natural language processing, mood tracking, and emergency response mechanisms to ensure that users can receive empathetic, data-driven mental health guidance while maintaining data privacy and emotional safety.

Key Components

- **User Authentication Module:** Enables users to create accounts securely or log in to existing ones, ensuring that personal data and mood history are accessible only to authenticated users.
- **Chat Interaction Module:** Allows users to engage with the AI chatbot through text-based conversations that provide emotional support, advice, and mental health tips using natural language understanding.
- **Breathing Exercise Module:** Provides guided breathing and mindfulness activities to help users manage anxiety and stress levels effectively.
- **Wellness Check & Mood Tracking Module:** Enables users to record daily moods and receive insights about their emotional patterns, encouraging proactive mental health care.
- **Crisis Contact Module:** Offers quick access to emergency contacts and mental health helplines when the system detects crisis-related phrases or high-risk indicators.
- **Insight and Analytics Module:** Displays user-specific insights, including emotional trends and recommendations for better well-being, ensuring continuous personal growth and self-awareness.

Use Case Diagram Overview

The AI Mental Health Platform is designed to connect three main users: Patients, Health Professionals, and Administrators each with distinct but complementary roles in supporting mental wellness and platform management.

Patients use the system as their personal mental health companion. They can create an account, start supportive conversations with the AI chatbot, practice guided breathing exercises to manage stress, and access immediate help through crisis support features. The platform also allows them to track daily moods and view personalized wellness insights, empowering them to better understand and manage their emotional wellbeing.

Health Professionals act as supervisors and care providers within the system. Their main tasks include reviewing patient activity dashboards, monitoring progress, and responding to crisis alerts when intervention is needed. This role provides a layer of professional oversight that complements the AI's supportive function, helping to maintain a safe and clinically informed environment.

Administrators oversee the overall operation of the platform. Their responsibilities include managing user accounts and permissions, ensuring the system runs smoothly, analyzing platform usage trends, and maintaining crisis logs for compliance and reporting. They also monitor technical performance and security to guarantee a reliable and confidential experience for all users.

Actor definitions

- **Patient:** A person seeking mental health support who interacts with the AI chatbot, tracks emotions, uses relaxation tools, and accesses crisis resources when necessary.
- **Health Professional:** A licensed mental health practitioner who supervises user wellbeing, monitors alerts, and provides remote clinical guidance.
- **Administrator:** A system manager responsible for user management, technical maintenance, and overall platform security.

Key use cases:

- Register and log into an account
- Start a therapeutic chat session
- Track mood and emotional patterns
- Practice breathing and relaxation exercises
- Access crisis or emergency support
- View personal wellness insights
- Monitor patient progress
- Review and respond to crisis alerts
- Generate progress reports
- Manage user accounts and permissions
- Analyze system and usage trends

- Monitor platform performance and reliability
- Update AI chatbot responses
- Maintain crisis intervention and compliance logs

In the corresponding use case diagram, each actor interacts only with the parts of the system relevant to their role. This structure clearly defines system boundaries, ensures secure access, and protects sensitive mental health data while maintaining a smooth and supportive user experience.

Descriptions	**Element**
Use case: • Must provide discernible value to an actor • Must form a complete conceptual function • Is contained inside the system's perimeter	
Actor: • Represent roles that humans, hardware devices, or external systems play while interacting with a given system • It may represent a user role performed by a system's contact with another system.	
A boundary: • It is a box that is drawn around the use case to depict the edge or boundary of the system. • Include the subject's name within or on top	Boundary
A Field relationship: • Connects an actor to the use case(s) it engages in	

Table 1: Symbols table

Use Case Diagram of the System

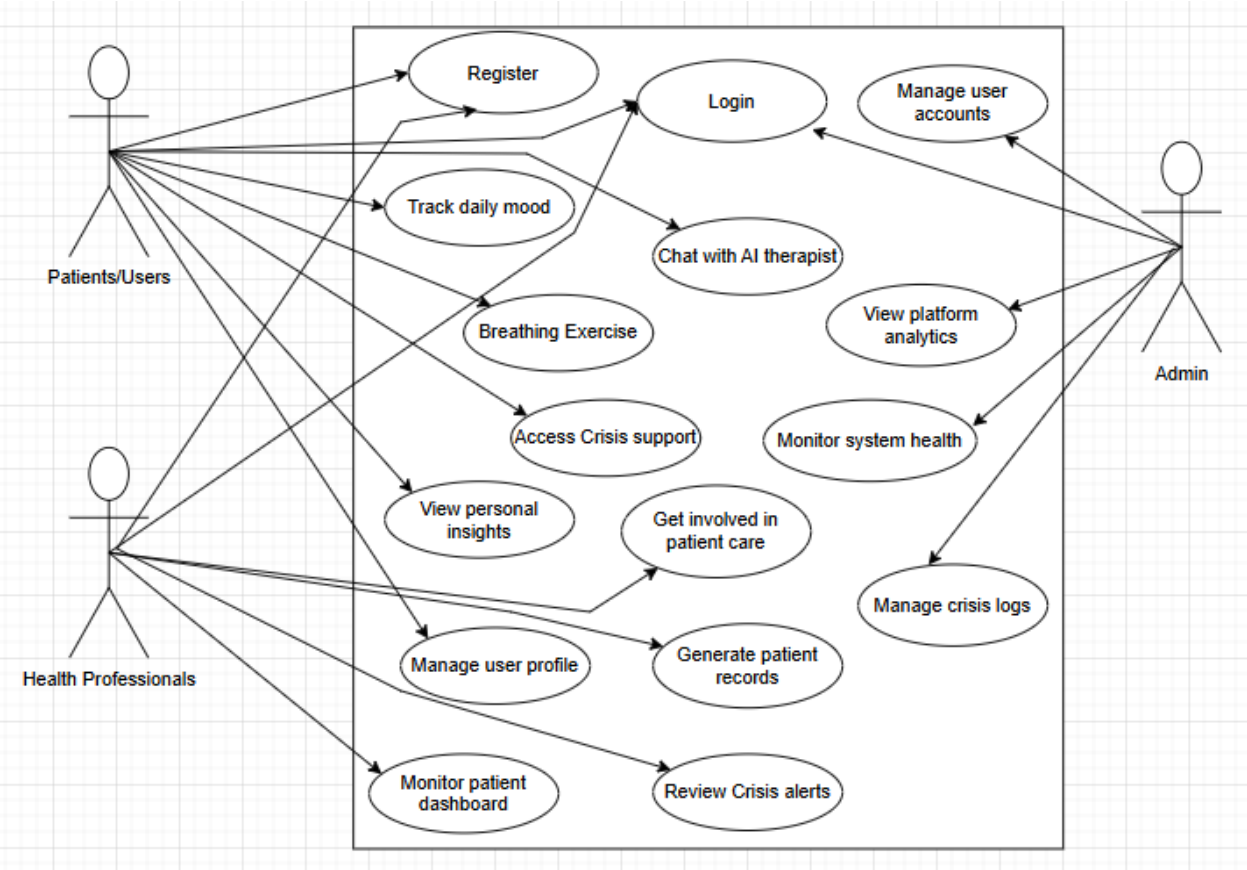

Figure 2: use case diagram

Use Case Descriptions

Use Case description specifies what a use case accomplishes, and what it requires to be properly performed. Each use case looks like this:

Name	Description
Actor	The participant in the use case
Pre-conditions	The system state is required before the use case may start
Post-conditions	When the use case is finished, the system states it
Normal flow	The use case's real steps
Alternative flow	Steps that could be taken if a regular flow were to fail

Table 2: use case description

Use Case 1: Sign Up/Login

Use case number:	**UC-01**
Use case name:	Register and login to account
Actor :	Patient, health worker, administrator
Description:	Allow users to create new accounts and securely log into system
Pre-conditions:	User must have valid credentials
Post-conditions:	User gains authenticated access to appropriate system based on role
Normal flow:	• User registers to the chatbot with required information • System validates information and creates account • User receives confirmation and can log in • System directs user to appropriate dashboard based role
Alternative flow:	• User forgets password and can reset it • User attempts to login with existing email/username

Table 3: Sign Up/ Login

Use Case 2: Start New Chat

Use case number:	UC-02
Use case name:	Initial chat session
Actor :	Patient
Description:	Patient to start conversation with the chatbot for assistant
Pre-conditions:	Must be authenticated
Post-conditions:	Chat session is created and available for review
Normal flow:	• Patient navigate to chat interface • System displays chat session or new chat option • Patient selects “Start new chat” • Patient begins conversation with mental health concerns • System provides empathetic responses and guidance • Chat history is saved
Alternative flow:	Patient resumes existing chat session

Table 4: Start New Chat

Use Case 3: Daily mood and emotions

Use case number:	UC-03
Use case name:	Track daily mood and emotions
Actor :	Patient
Description:	Allows patients to record and monitor their emotional states, mood patterns and affecting factors
Pre-conditions:	Patient must be authenticated
Post-conditions:	Mood entry is recorded
Normal flow:	• Patient selects mood tracking feature • System displays mood scale and emotional state options • Patient selects current mood level and identifies contributing factors • Patient adds optional notes about emotional context • System records mood entry
Alternative flow:	Patient skips optional notes and submits basic mood rating

Table 5: Track daily mood and emotions

Use Case 4: Crisis management

Use case number:	UC-04
Use case name:	Access crisis support resources
Actor :	Patient, health worker
Description:	Provides immediate access to emergency mental health resources and hotlines
Pre-conditions:	User identifies need for emergency support
Post-conditions:	User receives appropriate crisis resources and support options
Normal flow:	• User accesses crisis support selection • System displays emergency hotlines and immediate help options • User selects needed resource type • System provides contact information and access instructions • For patients: system notifies professionals of crisis access
Alternative flow:	User requires immediate emergency services contact

Table 6: Crisis Contacts Guidance

Use Case 6: Generate patient reports

Use case number:	UC-05
Use case name:	Generate patient progress reports
Actor :	Health professional
Description:	Allows health professionals to create comprehensive reports on patient progress
Pre-conditions:	Health professional must have patient data available for reporting
Post-conditions:	Report is generated and available for download
Normal flow:	• Health professional selects report generation feature • System displays report options • System processes patient data and generates report
Alternative flow:	Professionals generates reports for multiple patients

Table 7: Report generation

Use Case 7: Analyze chatbot trends

Use case number:	**UC-06**
Use case name:	Analyze chatbot usage trends
Actor :	Admin
Description:	Allows admin to review chatbot-wide usage patterns and system performance status
Pre-conditions:	System must have sufficient usage data for meaningful trends
Post-conditions:	Admin gains insights into system performance and user engagement
Normal flow:	Admin accesses analytics dashboard Admin displays system health and status
Alternative flow:	Admin generates system health reports

Table 8: System health reports

Class Diagram

The class diagram for the AI-Based Mental Health Support System provides a clear overview of the platform's core structure and interactions between its main components. It highlights key classes such as User**,** Patient**,** Health Professional**,** Administrator**,** Chat Session**,** Mood Entry**,** and Crisis Support, along with their essential attributes and operations. Relationships among these classes show how users interact with the system from account registration and mood tracking to chatbot conversations and emergency interventions.

Overall, the diagram illustrates how each class contributes to delivering personalized mental health support, maintaining secure data handling, and ensuring coordinated collaboration between patients, professionals, and administrators within the platform.

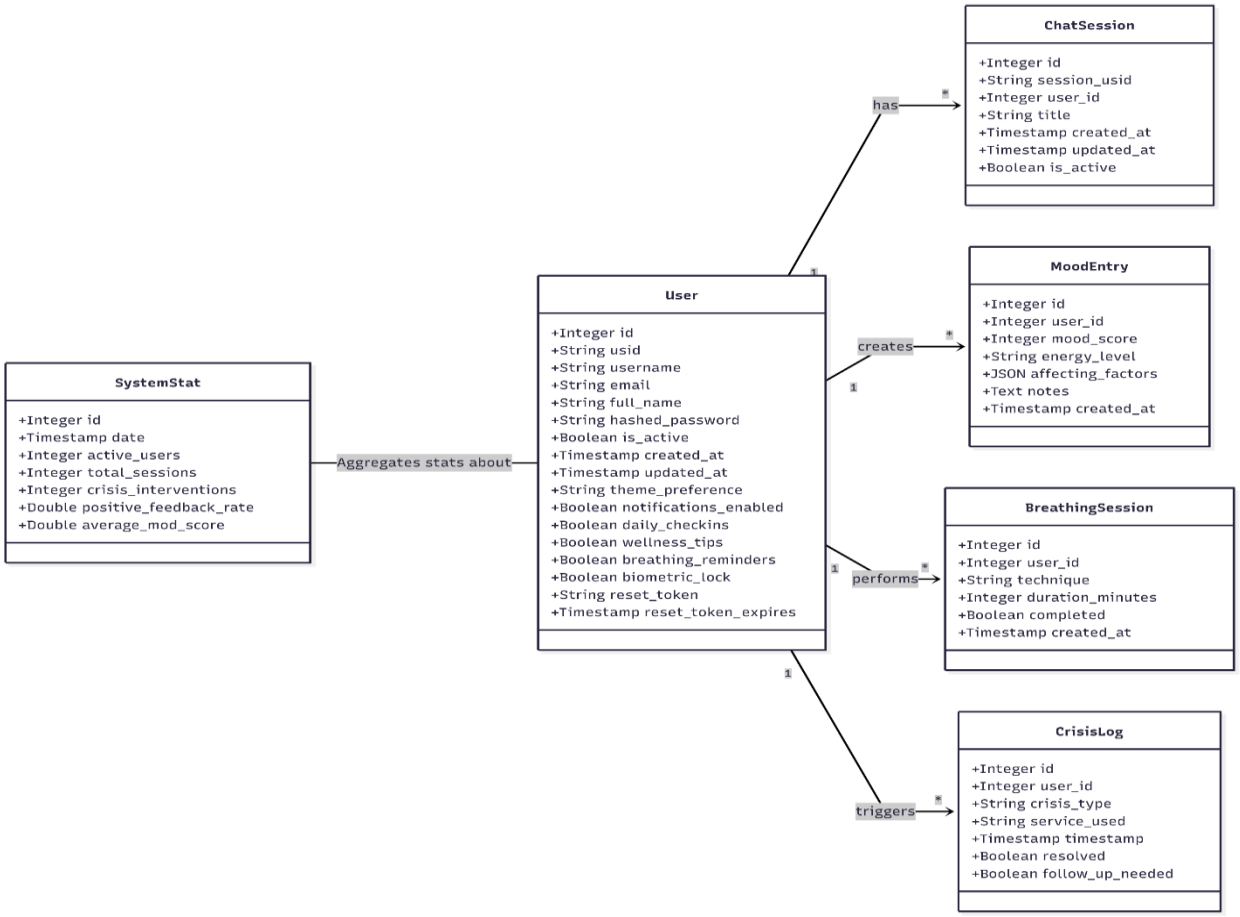

Figure 3: class diagram

Sequence Diagram

A UML Sequence Diagram is a visual tool used to show how different parts of a system interact with each other over time. It focuses on the flow of messages between users and system components, helping to illustrate the sequence in which events occur to complete a process or fulfill a use case. The diagram's vertical axis represents time, showing when interactions begin, how they progress, and when they end making it easy to follow the logic and timing of operations within the system.

In the context of the AI Mental Health Support System, sequence diagrams play an important role in visualizing how the system behaves in real-life situations. They show the step-by-step flow of communication between the user (patient), the AI chatbot, health professionals, and administrative components. This helps clarify how different system elements work together to provide responsive and secure mental health support.

By visualizing these interactions, sequence diagrams help developers, designers, and mental health professionals understand how system components coordinate to deliver care safely and efficiently. They ensure that all interactions are properly timed, respect user privacy, and align with established therapeutic standards.

Sequence diagram: patient operations

This figure illustrates the workflow for a patient using an AI chat system. The process begins with a login request, where the Auth API validates the patient's credentials against the database. Once authenticated, the patient can start a chat, initiating the "Daily Share" process, during which the Chat Service creates a new session. When the patient sends a message, it is routed to the AI Engine for processing and generating a response. The response is then returned through the Chat Service, which records the interaction and displays both the patient's message and the AI's reply.

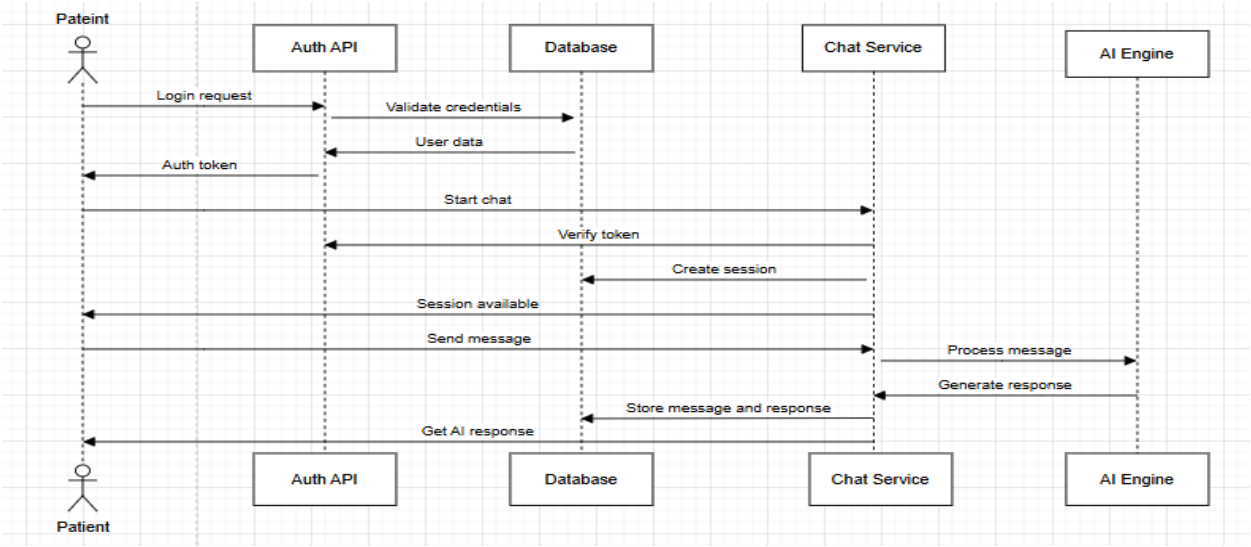

Figure 4: Patient interaction process

Sequence diagram: health professional

This diagram illustrates the workflow for a professional, such as a clinician, accessing the monitoring system. The process begins with a login request, where the Auth API verifies their credentials and role against the database. Upon successful authentication, the professional can access their data and request the patient list to view an overview. The system also includes a Monitoring Service that generates notifications and alerts, which are managed by an Alert Service to keep the professional informed.

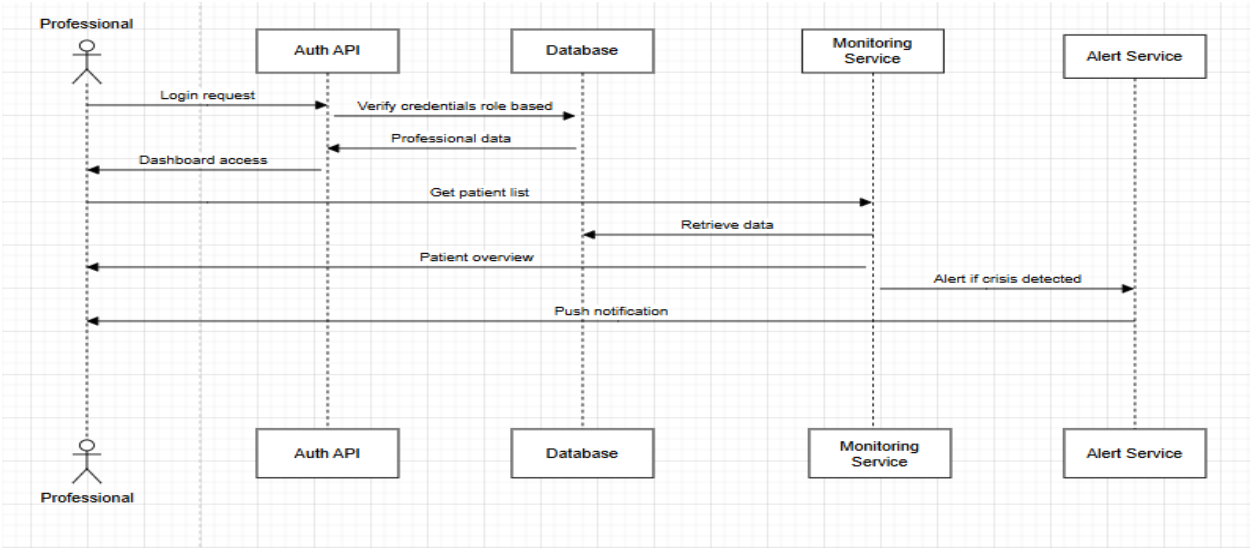

Figure 5: health professional

Sequence diagram: admin main operations

This diagram outlines the system access and administrative workflow for an Admin user. The process begins with a login request, where the Auth API verifies the user's admin privileges against the database. Once access is granted, the admin can perform actions such as querying the users' table and requesting a list of all users and their roles through the User Management Service. Additionally, the admin can use the Analytic Service to generate reports, consolidating data for oversight and analysis.

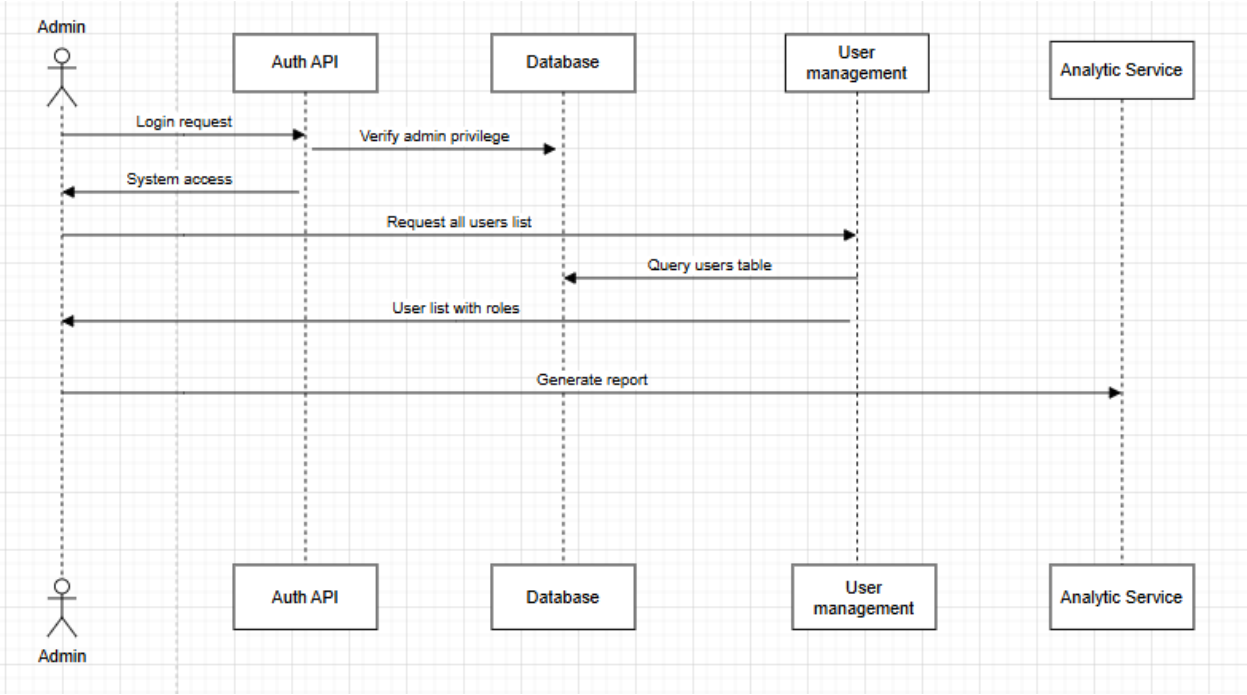

Figure 6: admin main operations

Activity Diagrams

In the AI-Based chatbot for mental health support, the activity diagram models the flow of user interactions and system responses across core functionalities of the platform. It visually represents the dynamic sequence of activities, helping developers and stakeholders understand how users move through the system. The diagram starts with a single initial node—typically when a user accesses the website or app. From there, it maps out the main activities a user can perform, such as:

- **User Registration**: Enter details → Validate input → Store in database → Confirm success.
- **Mood Tracking**: Select mood → Add notes → Save log → Update mood dashboard.

This diagram is a key tool for analyzing user behavior, identifying flow inefficiencies, and optimizing the user journey on the chatbot platform. It supports a more intuitive design and improved system performance by mapping real user paths.

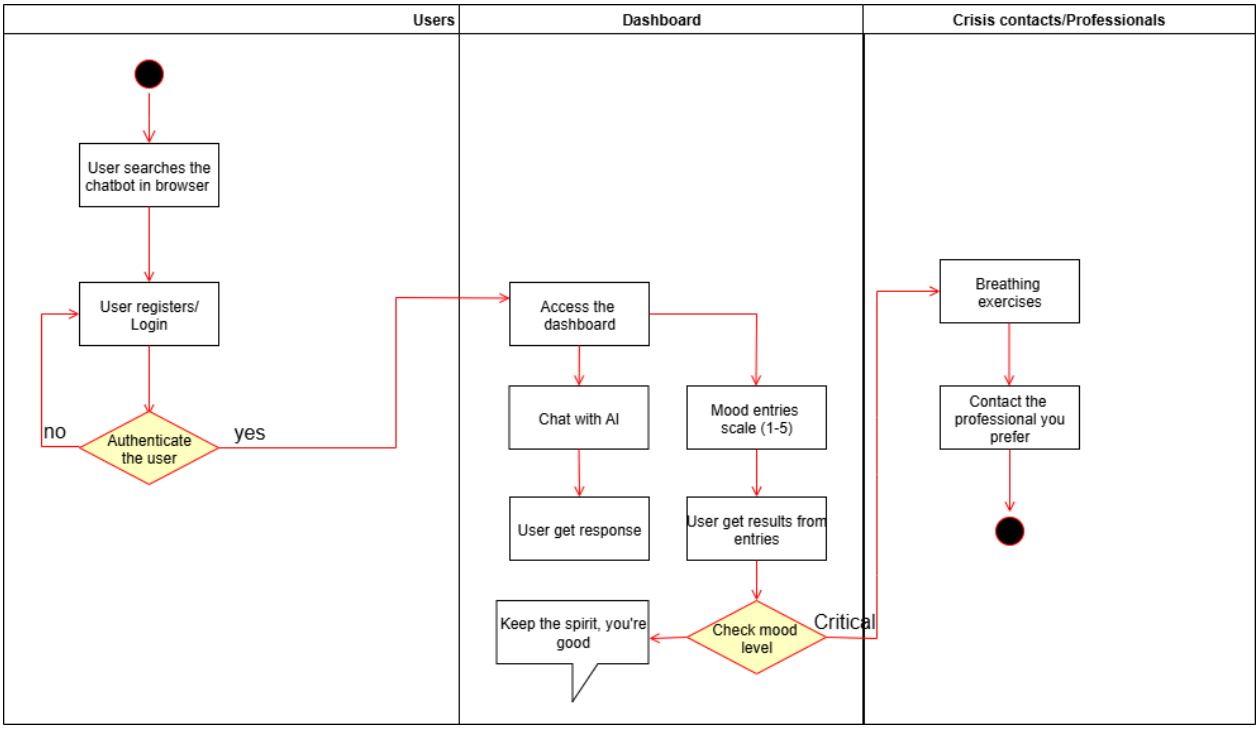

Figure 7: activity diagram

Database Design and Data Dictionary

The database design for the AI chatbot for mental health support is built on a structured relational model that focuses on both user management and therapeutic interaction. At its foundation are user-related tables that manage authentication and enforce role-based access for patients, health professionals, and administrators, ensuring that each user interacts with the system according to their designated privileges.

The therapy-centered components handle key areas such as chat sessions**,** message history**,** mood tracking, and breathing exercise records. These elements work together to provide continuity of care, enabling users and professionals to review progress and maintain consistent engagement over time.

In addition, crisis management tables play a vital role in recording emergency incidents, generating alerts, and supporting structured escalation protocols to ensure timely intervention when needed. Supporting structures also include assignment and activity logs, which map patients to health professionals and track user actions for transparency and accountability.

Altogether, the database forms a reliable, efficient, and secure foundation for delivering high-quality digital mental health services.

System Architecture Design

The AI chatbot architecture for mental health support is designed to provide a safe and supportive digital space. The frontend, built with Next.js and Tailwind CSS, offers a simple, welcoming interface for chatting with the AI, tracking moods, and practicing exercises. The backend, powered by FastAPI, handles authentication, analyzes emotional tone, and flags potential crises for human support. All user data, including chat sessions, mood entries, and alerts, is securely stored in a PostgreSQL database, ensuring privacy and continuity of care.

This diagram outlines the core features and user flow of a mental health support system:

The application begins with user login or registration, managed by the Auth API and database. Once authenticated, the user accesses the main dashboard, which includes an AI chat for support,

wellness tips, guided breathing exercises with animations, and a list of critical contacts for urgent situations. The workflow for each feature shows the user's actions on the frontend and the corresponding API requests and data exchanges with backend services to validate tokens, send messages, fetch content, and log user sessions.

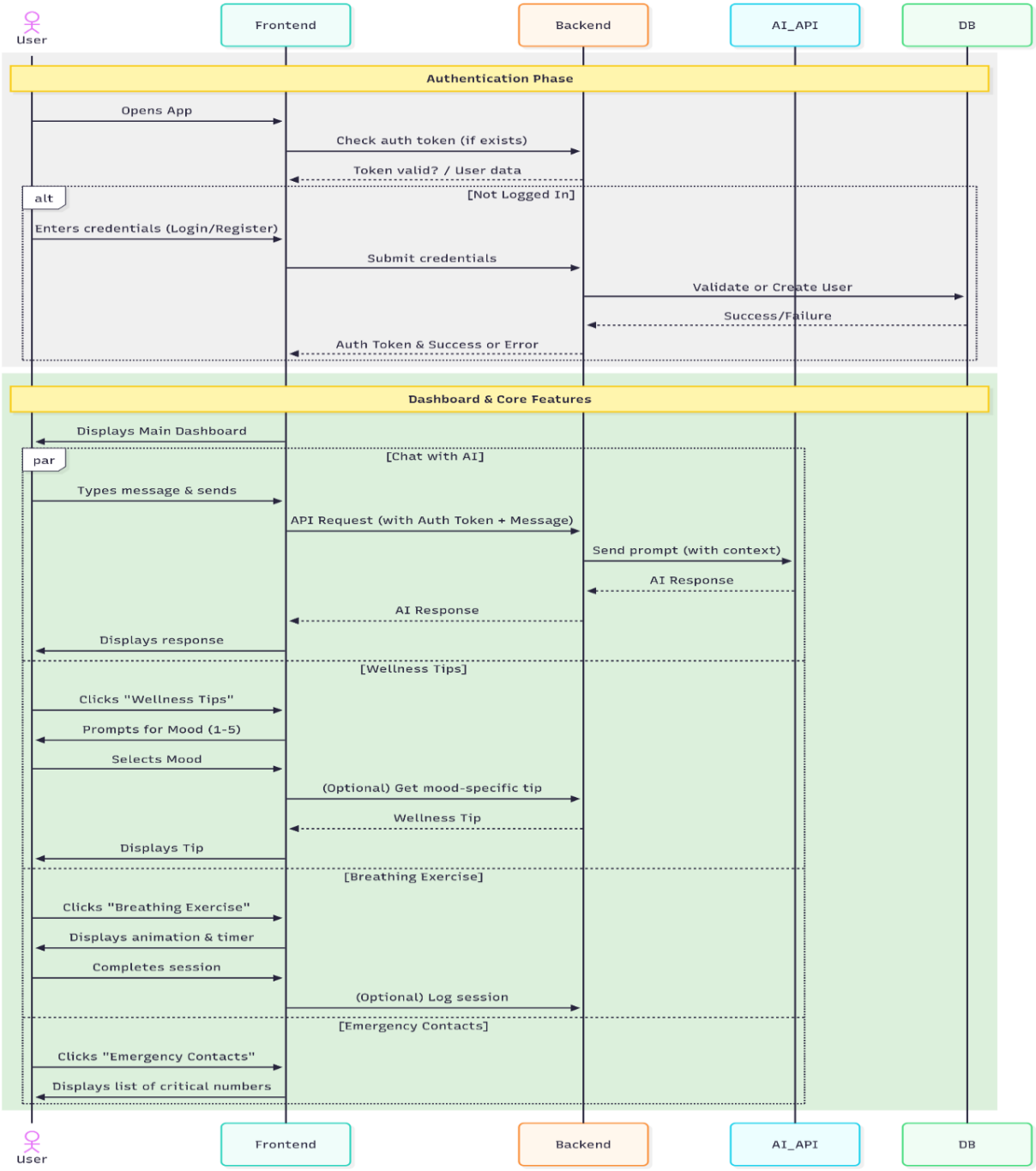

Figure 8: system architecture design

CHAPTER 4

IMPLEMENTATION OF THE NEW SYSTEM

Introduction

This chapter presents the journey of developing and deploying the AI-based chatbot for mental health support, a system designed to blend technology with compassion. Building on earlier chapters that outlined requirements, architecture, and design, this section brings the platform to life. It explains the technologies and tools that power the system, showcases its main user interfaces, and highlights the testing, hardware, and software components that make the system secure, reliable, and ready for real-world use. Additionally, it discusses how user feedback was incorporated during development, how data privacy is maintained, and how the system ensures timely and personalized mental health support for users.

Technologies Used:

Frontend

The frontend represents the face of the AI based chatbot for mental health support the part users see and interact with. It was built with modern, flexible tools to ensure a smooth and supportive experience for anyone seeking help.

- **Next.js:** was used to create a fast, server-rendered web application that allows seamless page transitions and responsive interactions. This ensures that users can move effortlessly between features such as therapeutic chats, mood tracking, and personal dashboards.
- **Tailwind CSS:** helped design a clean, calming, and fully responsive interface. Its utility-first approach allowed for consistency and accessibility across devices, so users can engage comfortably on phones, tablets, or computers.
- **TypeScript:** added an extra layer of reliability by catching coding errors early and improving performance, ensuring the platform runs smoothly and securely.
- **Visual Studio Code:** served as the main workspace for writing, testing, and managing the application's code, supporting efficient development and debugging.

Backend

Behind the scenes, the backend serves as the intelligent core of the AI based chatbot for mental health support handling user data, managing conversations, and ensuring everything runs efficiently.

- **FastAPI (Python):** powers the system's logic and API endpoints. It handles user authentication, manages chat sessions, and processes data used for analytics.
- **PostgreSQL:** acts as the platform's secure database, storing sensitive information such as user profiles, mood entries, chat histories, and crisis logs safely and efficiently.
- **SQLAlchemy ORM:** bridges the backend and database, making data interactions secure and seamless without the need for complex manual queries.
- **JWT:** ensures secure and reliable user sessions, keeping data protected and maintaining trust between users and the system.
- **AI & Specialized Services**: The heart of AI based chatbot for mental health support lies in its intelligence the ability to understand, respond, and provide meaningful support.
- **AI/NLP Engine** gives the chatbot its empathetic voice. It interprets messages, detects emotional tones, and generates supportive, context-aware responses. When necessary, it identifies crisis-related language to trigger human intervention.
- **Notification Service** enhances responsiveness by instantly alerting health professionals to emergencies and sending timely updates or reminders to users.

Together, these technologies work harmoniously to create a platform that feels human-centered, trustworthy, and responsive an always-available companion for mental health support.

Presentation of the New System

Home page:

This landing page is the first impression when visiting the chatbot and it is designed to make you feel calm and understood right away and it gently shows you how the app can be helpful companion as someone to talk to, a way to track your mood, and a source of support if things get tough.

Everything from the soft colors to the encouraging messages that makes you feel safe and hopeful

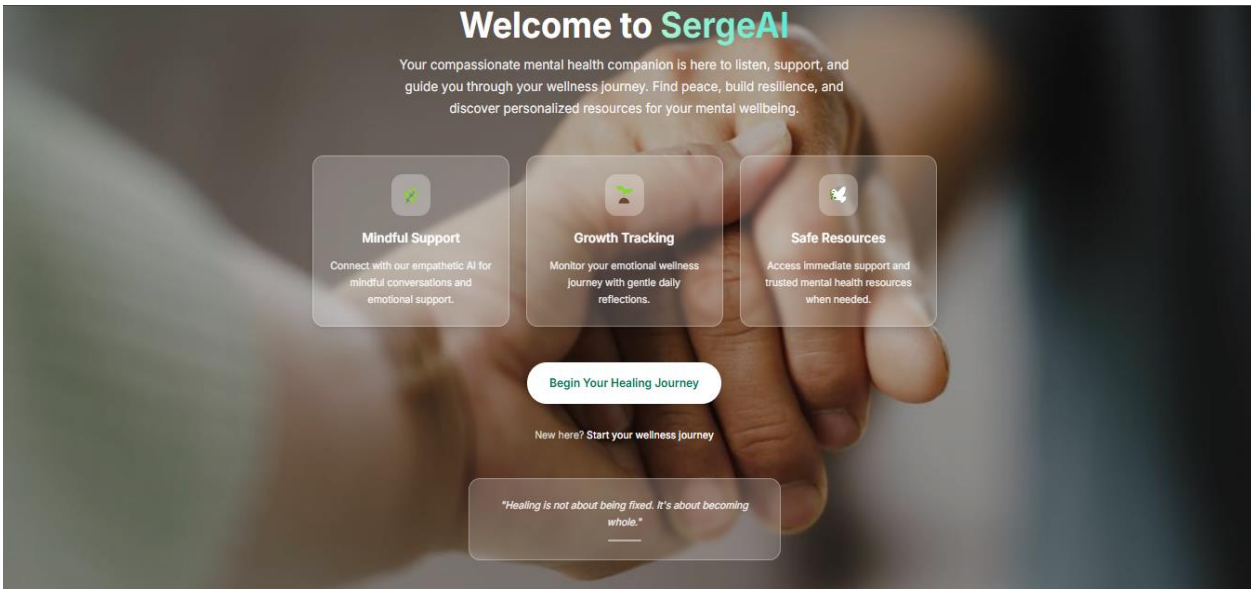

Figure 9: landing page

The Login page:

The login page acts as a familiar and secure checkpoint for returning users. It's designed for simplicity and speed, providing a straightforward form for username or email and password so you can seamlessly pick up where you left off and the reset password option whenever the user forgets their password.

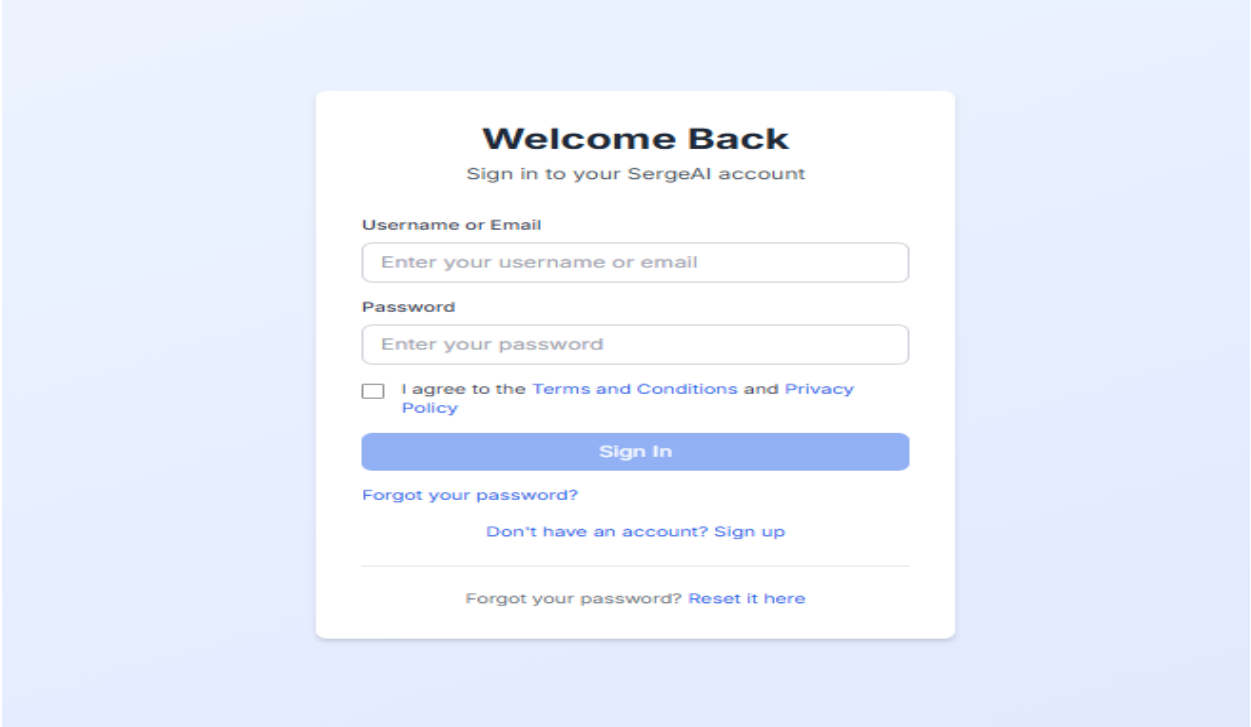

Figure 10: login page

The Forgot password page:

The forgot Password page provides a simple and user-friendly way for someone to reset their password. It shows a clean form where the user enters their email address, then clicks “Send Reset Link” to receive a password-reset email. There is also a “Back to Login” link that allows users to return to the login page if they remember their password or accessed the page by mistake. The design is minimal, clear, and focused on helping the user regain access quickly.

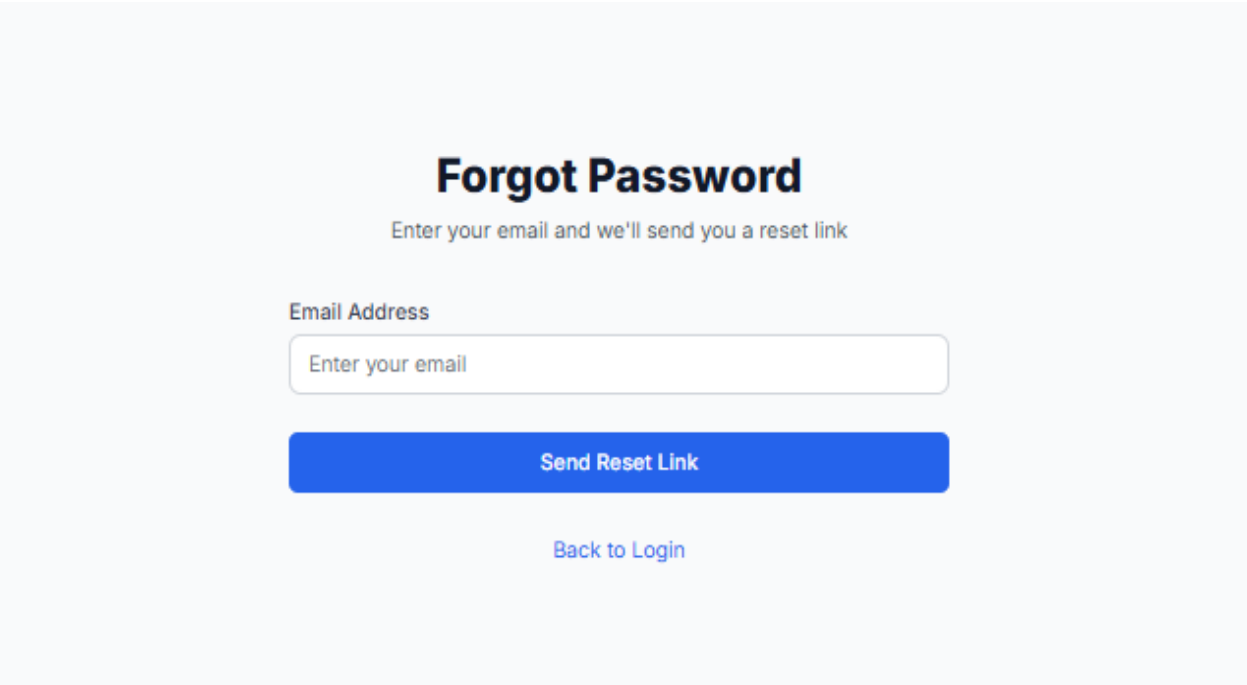

Figure 11: reset password page

Sign up page:

In contrast, the signup page serves as a welcoming introduction for new users. Its purpose is to warmly onboard you by collecting just enough information like your name, email and a new password to create your personal account. This page is built on a foundation of trust, reassuring that your information is safe, and clearly communicating the value of taking this first step to begin your journey

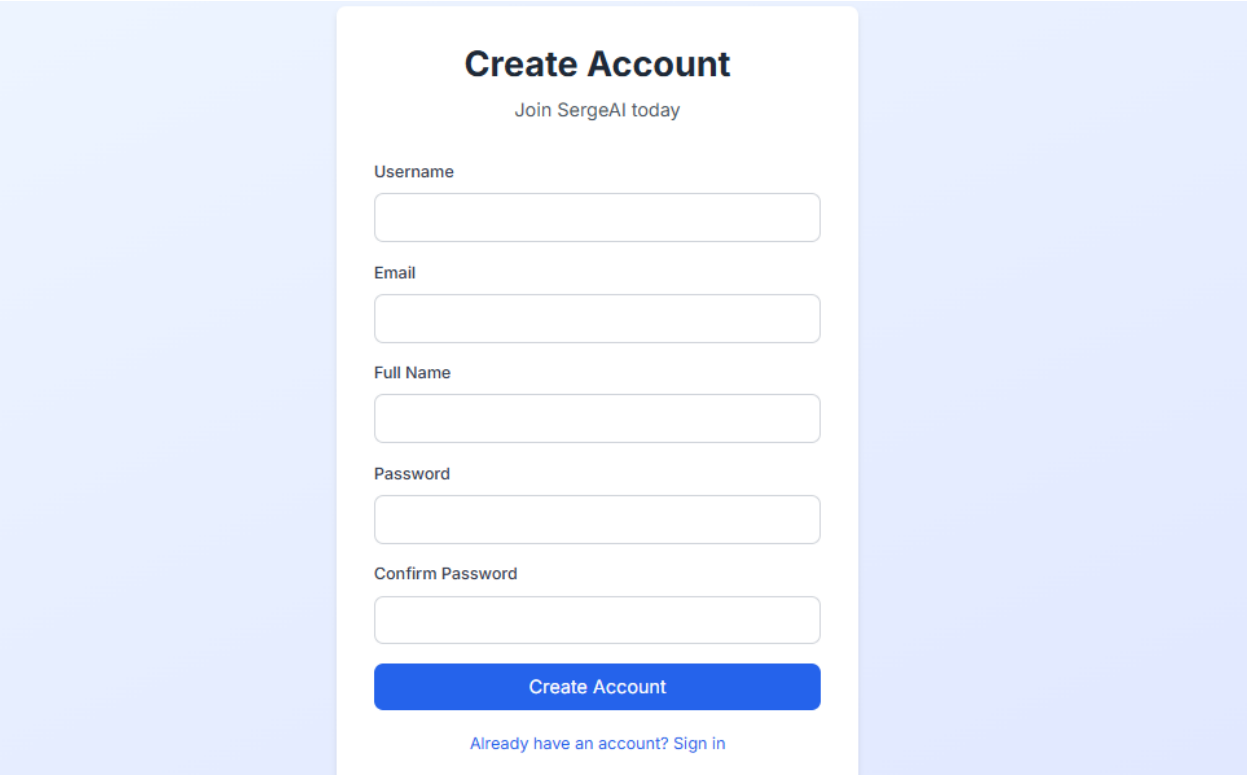

Figure 12: sign up page

Patient Dashboard:

Is the heart of an individual's mental wellness experience on the AI based chatbot for mental health support, It's designed to feel warm, personal, and supportive like a safe digital space where users can reflect, heal, and grow. Here, individuals can easily view their recent activities, including past chat sessions, mood logs, and completed breathing or relaxation exercises. The layout is intentionally simple and soothing, helping users focus on their well-being without feeling overwhelmed. With just a click, they can start a new conversation with the AI therapist, record how they're feeling, or begin a calming breathing session whenever they need a moment of peace.

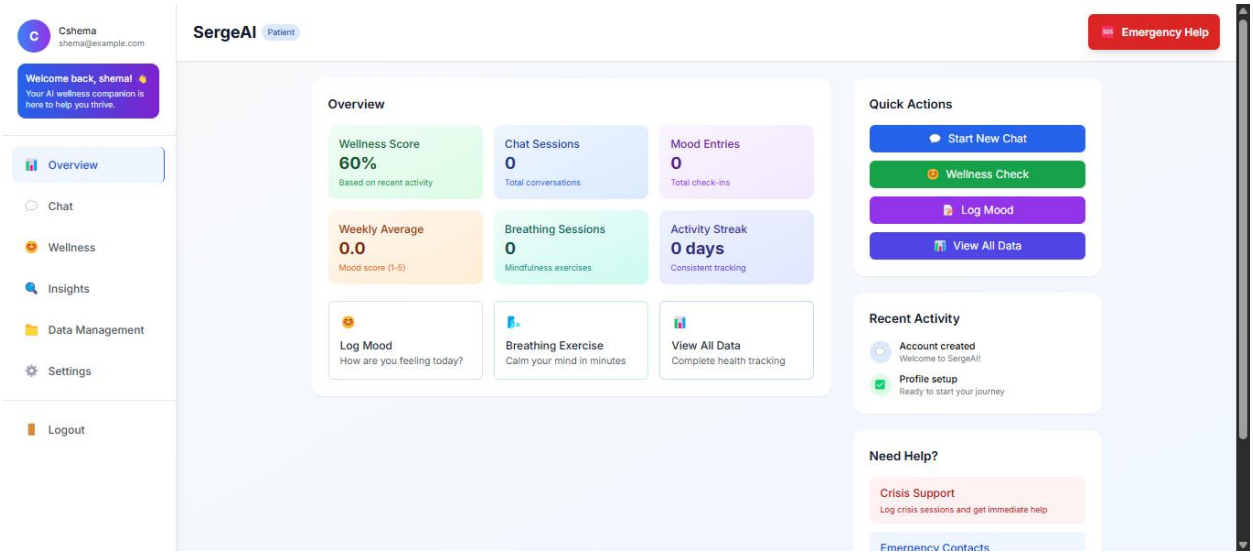

Figure 13: patient dashboard

AI Chat Interface:

Is the core feature of the AI based chatbot for mental health support, offering users a safe and empathetic space to express their thoughts and emotions. It enables real-time, supportive conversations with the AI therapist, which listens, understands emotional cues, and responds with compassion. Designed for comfort and clarity, the chat interface ensures every interaction feels natural, private, and reassuring providing immediate mental health support anytime it's needed.

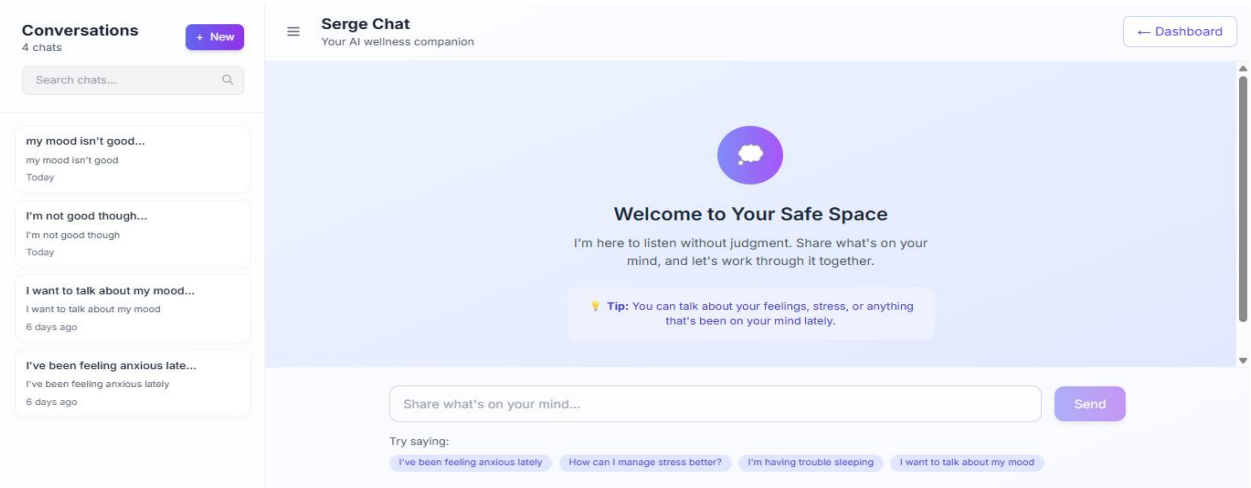

Figure 14: AI chat interface

Data Management:

Data management tab acts as your personal wellness archive, giving you a simple and secure way to reflect on your mental health journey. It lets you revisit past chat sessions, track mood changes over time, and view your breathing exercise history all in one place.

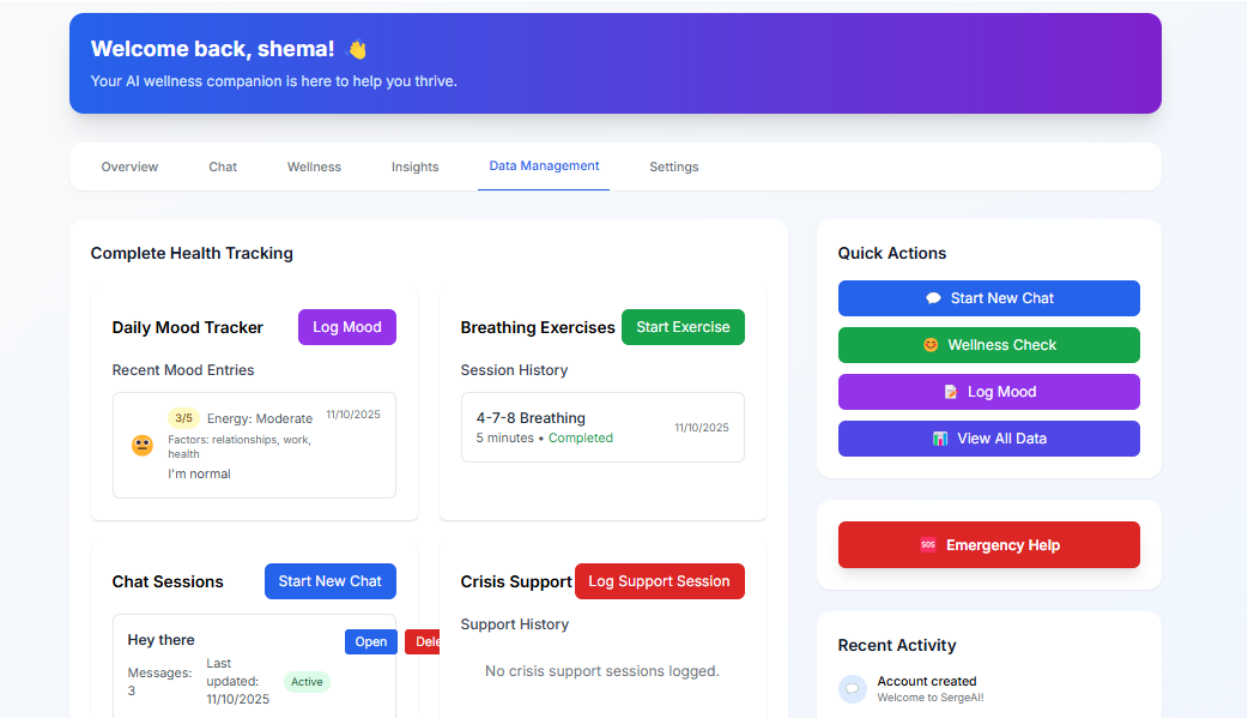

Figure 15: data management tab

Health Professional Dashboard:

The professional dashboard serves as a centralized command center for managing patient care. It presents essential insights such as active patient counts, current crisis alerts, and recent mood trends at a glance. This dashboard allows professionals to efficiently monitor overall wellbeing, quickly identify individuals who may need urgent support, and access detailed patient histories to provide informed and compassionate interventions.

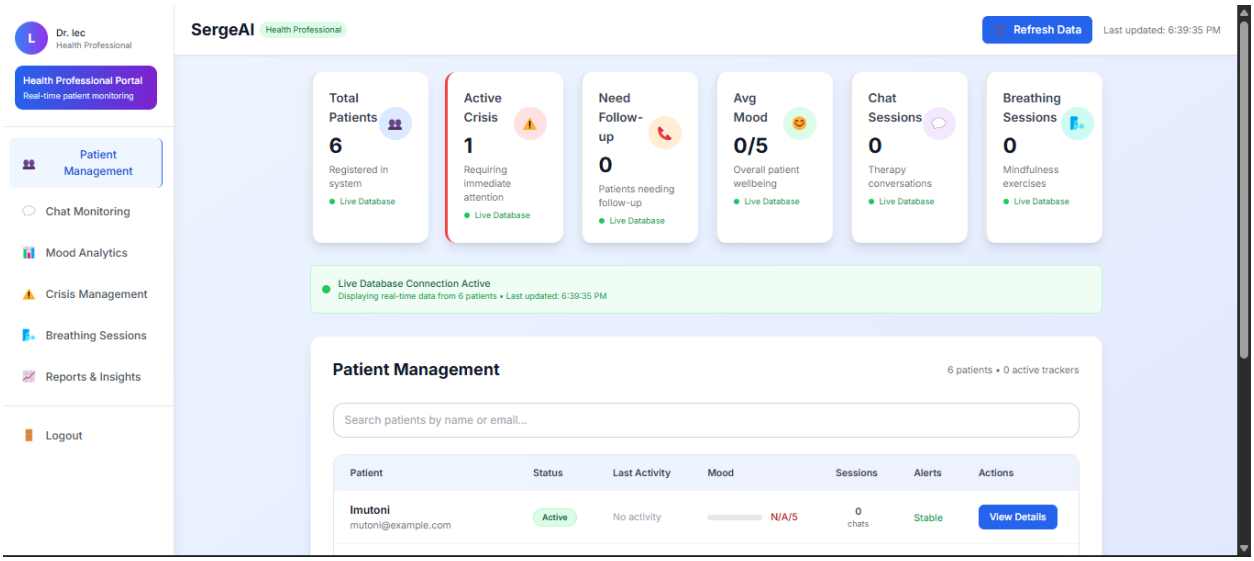

Figure 16: health professional dashboard

Admin Dashboard:

The admin dashboard offers a centralized view of the platform's overall performance, displaying key insights such as user statistics, system status, and ongoing activities. It allows administrators to manage user accounts, monitor analytics, and configure system settings to maintain smooth and secure operations across the platform.

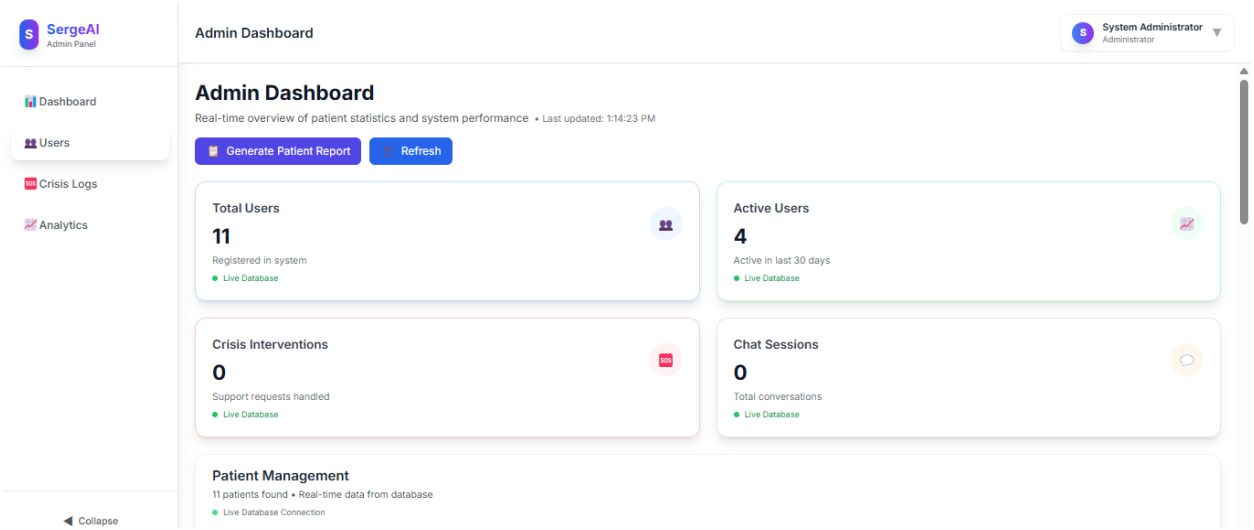

Figure 17: admin dashboard

User Management:

The User Management section displays a structured list of all system users, including their names, email addresses, account status, assigned roles, and the dates they joined the platform. Each user entry includes quick-action controls Activate, Deactivate, and Delete allowing the administrator to manage user accounts efficiently. This interface provides a clear, organized view that supports effective monitoring and control of user access within the system.

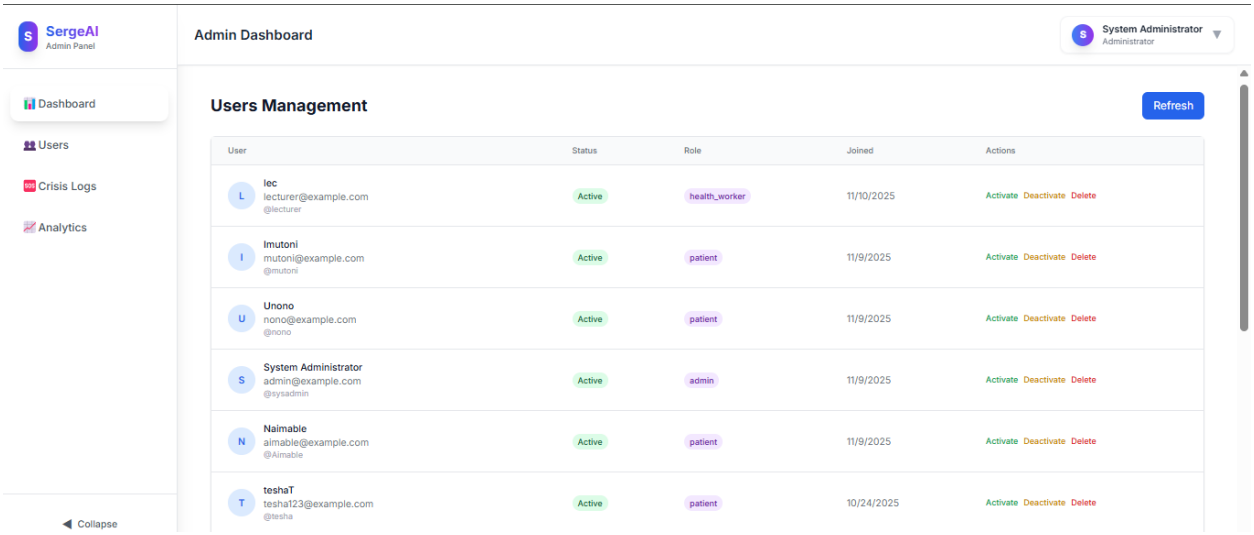

Figure 18: user management

Software Testing

Software testing played a crucial role in ensuring the reliability and quality of the AI-based chatbot for mental health support. It was carried out to make sure the system worked as intended, met user needs, and delivered a smooth, safe, and supportive experience. Through this process, potential issues were identified and resolved early, improving both performance and user trust in the final product. Testing also helped verify the system's responsiveness, data security, and overall usability, ensuring it could handle real-world scenarios effectively.

The testing process focused on verifying that all components aligned with design standards, functioned properly, and met stakeholder expectations. The main testing phases included the following:

Unit Testing:

Each part of the system was tested individually to confirm that it performed its role correctly. This involved checking the user authentication module for secure login and role management, ensuring the mood tracking feature accurately recorded users' emotional data, testing the chat handling system for proper message delivery and session flow, and validating the crisis detection algorithms to correctly recognize signs of distress. This step-by-step testing helped confirm that every feature was dependable before integrating it with others.

Integration Testing:

Once individual components were verified, integration testing ensured they worked well together. This stage tested the communication between the frontend and backend APIs, the coordination between the chat service and AI engine for generating empathetic responses, and the connection between the application and database to ensure secure, consistent data storage. Special attention was given to the crisis alert process, confirming that when a potential emergency was detected, alerts were accurately triggered and health professionals were promptly notified. This phase also validated overall system stability under different user scenarios and confirmed that all integrated features provided a seamless and reliable user experience.

Validation Testing:

The system validation testing thoroughly evaluated the entire platform against its original goals and user expectations. It involved checking that the interface was easy to navigate and accessible, ensuring that the conversation logic and mood tracking provided real therapeutic value to users, and testing the crisis alerts and notifications to confirm they functioned effectively during emergencies. In addition, data security and accuracy were carefully audited to maintain the integrity of all stored information, while performance tests were conducted to verify that the system remained stable and responsive even under heavy usage conditions.

Overall, the testing process confirmed that AI based chatbot for mental health support is a reliable, efficient, and secure mental health support system. It delivers a responsive, user-friendly, and emotionally supportive experience for individuals seeking guidance and reassurance.

Ethical and Privacy Considerations

The mental health chatbot is designed with privacy and ethical responsibility at its foundation. Users are able to interact with the system anonymously, with no requirement for personally identifiable information such as names or phone numbers. For optional features like mood tracking, pseudonymous accounts are sufficient. Data collection is kept to a minimum, focusing only on chat interactions and mood entries, and excluding unrelated information like location or contacts. Any data used for training and improvement is anonymized to protect individual identities.

Transparency is central to the project. A clear privacy policy outlines what data is collected, the reasons for collection, how it is secured, and the limited cases in which anonymized data may be shared. Users provide informed consent before any data is gathered, ensuring they understand the role and limits of the chatbot. The system also makes clear that it is an AI assistant offering supportive resources rather than a licensed therapist.

Although Rwanda does not currently have the same legal frameworks as the GDPR or HIPAA, the chatbot aligns with their core principles. It ensures fairness, lawfulness, and purpose limitation, guaranteeing that all collected data is used solely for supporting mental health and not for unrelated purposes such as marketing.

Terms and privacy document:

The Terms of Service: clarify that the service is a preliminary support tool and not a substitute for professional healthcare. It outlines user responsibilities, limits the company's liability, and specifies conditions for account termination.

The Privacy Policy: describes the personal information collected, including account details, chat conversations, and mood entries. It explains that this data is used to provide and improve the service, personalize the user experience, train AI models with anonymized data, and maintain platform security.

Legal

Last updated: 9/20/2025

Terms of Service

1. Nature of Service

SerenityBot is an AI-powered tool designed to provide preliminary mental health support and resources. It is not a substitute for professional medical advice, diagnosis, or treatment. Always seek the advice of qualified healthcare providers with any questions you may have regarding medical conditions.

2. User Responsibilities

You agree to use our services responsibly and not for any unlawful purposes. You are responsible for maintaining the confidentiality of your account credentials and for all activities that occur under your account.

3. Limitation of Liability

SerenityBot and its developers are not liable for any direct, indirect, incidental, or consequential damages resulting from your use or inability to use our services, including reliance on any information provided by the AI chatbot.

4. Account Termination

We reserve the right to suspend or terminate your account if you violate these terms or engage in activities that may harm our services or other users.

Privacy Policy

1. Information We Collect

We collect information you provide directly, including:

- Account information (username, email)
- Chat conversations with our AI assistant
- Mood tracking entries and self-assessments
- Technical information (IP address, browser type)

2. How We Use Your Information

Your information is used to:

- Provide and improve our mental health support services
- Personalize your experience
- Train and enhance our AI models (with anonymized data)
- Ensure platform security and prevent misuse

3. Data Security

Figure 19: terms and privacy page

Hardware and Software Requirements

Client-Side Software Requirements:

Operating System: Compatible with Windows 10 or later, macOS 10.14 or later, and popular Linux distributions.

Mobile Operating System: Compatible with Android 8.0 or later and iOS 18.0 or above.

Web Browser: Supports the latest versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari (macOS).

Client-Side Hardware Requirements

Processor: Minimum dual-core processor (Intel i3 or AMD equivalent), with a quad-core processor (Intel i5 or AMD equivalent) recommended.

Mobile Processor: Minimum quad-core processor (ARM Cortex-A53 or equivalent), with an octa-core processor recommended.

RAM: At least 4 GB, with 8 GB or more recommended.

Mobile RAM: At least 8 GB, with 16 GB or more recommended.

Storage: Minimum of 256 GB HDD or SSD, with 512 GB SSD preferred.

Mobile Storage: Minimum of 16 GB, with 32 GB or more recommended.

Display: Minimum resolution of 1280 × 720, with 1920 × 1080 or higher recommended.

Mobile Display: Minimum resolution of 720 × 1280, with 1080 × 1920 or higher recommended.

Network Connection: Stable internet connection with a minimum download speed of 5 Mbps, with 10 Mbps or higher recommended.

Peripherals: Keyboard, mouse or trackpad, speakers or headphones, webcam, and microphone for communication and collaboration.

Mobile Peripherals: Touchscreen, speakers or headphones, and microphone.

Server-Side Software Requirements

Backend Framework: Developed using FastAPI, running on Python 3.8 or later.

Database: Uses PostgreSQL 16 as the primary database, integrated through SQLAlchemy ORM.

AI/ML Processing: Implements Python-based NLP frameworks such as spaCy and Transformers.

Authentication: Utilizes JWT for secure and scalable user authentication.

Frontend Deployment: Hosted on Vercel for optimized web delivery.

Backend Hosting: Deployable on major cloud platforms such as AWS, Google Cloud, or Microsoft Azure.

Server-Side Hardware Requirements

Processor: High-performance, multi-core processor (quad-core recommended).

RAM: At least 4 GB, with 8 GB or more recommended for production.

Storage: Minimum of 20 GB SSD storage, with additional capacity for database growth.

Network Connection: Reliable high-bandwidth internet connection with a minimum throughput of 100 Mbps.

Backup: Automated regular backups for database and user data to ensure data security and recovery.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

Conclusion

The AI-Based Mental Health Support System effectively addresses key gaps in mental healthcare accessibility by providing a technology-driven solution for emotional support and wellbeing monitoring. Traditional mental health services often face challenges such as limited accessibility, long waiting times, and stigma associated with seeking help. This system overcomes these barriers by offering immediate, personalized support through conversational AI, mood tracking, and crisis intervention, ensuring users can access care regardless of location or time constraints.

The platform's development emphasized user-centered design, data security, and ethical AI practices. Personalized interactions allow the chatbot to provide empathetic, context-aware responses, while continuous mood monitoring helps users understand their emotional patterns and develop self-awareness. Privacy and confidentiality were prioritized, ensuring sensitive information remains protected. By combining NLP with mental health expertise, the system delivers reliable, supportive, and human-like interactions, complementing traditional mental health services.

This project demonstrates how AI can enhance the reach and effectiveness of mental health care. Its modular and scalable design allows for future expansions, including integration with clinical services, support for multiple languages, and new accessibility features. By providing a safe, stigma-free environment for emotional support, the system contributes significantly to the growing field of digital mental health technologies and offers a viable solution for bridging the global mental health support gap.

In summary, the AI-Based Mental Health Support System represents a practical, ethical, and scalable tool that empowers individuals to manage their mental wellbeing while supporting the broader goal of accessible, inclusive mental health care. Its continued development and adoption can improve lives, foster resilience, and help build a healthier society.

Recommendations

To Ndera Neuropsychiatric Teaching Hospital:

The AI-Based Mental Health Support System represents a significant step toward digital innovation in mental health care. To maximize its benefits, stakeholders should focus on:

Conducting regular training sessions for health professionals to ensure effective use of the monitoring dashboard and crisis intervention features.

Establishing clear protocols for escalating critical cases from the AI system to human professionals.

Developing comprehensive data governance policies to ensure compliance with healthcare regulations and protect patient privacy.

Expanding system capabilities to include broader language support and culturally relevant responses to better serve diverse user groups.

Developing mobile applications for iOS and Android to improve accessibility and user engagement.

Integrating voice-based interactions to accommodate users with different preferences and abilities. Implementing advanced analytics to predict mental health trends and support timely interventions.

To the Future Research:

While the system enhances mental health support accessibility and efficiency, further improvements can enhance functionality and long-term impact:

Conducting longitudinal studies to assess the long-term outcomes of AI-supported mental health interventions.

Developing specialized modules for specific mental health conditions and demographic groups.

Investigating federated learning approaches to improve AI models while maintaining user privacy.

Establishing partnerships with healthcare institutions for clinical validation and integration into existing care pathways.

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APPENDICES

CURRICULUM VITAE

Mupenzi Serge

sergemupenzi2001@gmail.com

+250 787 953 559
Kigali, Rwanda

SKILLS

Technical Skills:
Java, C#, Python, JavaScript, Spring Boot, ASP.NET Core, RESTful APIs, PostgreSQL, SQL Server, HTML5, CSS3, Bootstrap, Tailwind, Power BI, Git/GitHub, Figma, CyberSecurity, Networking

Soft Skills:
Problem-solving, teamwork, adaptability, and analytical thinking

Leadership & Communication:
Collaborative leadership in project teams, mentoring peers, and effective technical communication

WORK EXPERIENCE

IT Intern – Bank of Kigali, Kigali, Rwanda
March 2025 – September 2025
• Resolved technical support tickets, cutting average response time
• Assisted in deploying system updates and security patches
• Supported infrastructure optimization and cybersecurity compliance initiatives

EDUCATION

Bachelor of Science in Information Technology (Software Engineering)
Adventist University of Central Africa, Kigali —awaiting graduation
Relevant Coursework: Software Engineering, Database Management, Data Structures, Web Development, Network Security

CERTIFICATIONS

• Data Analysis Fundamentals — Udacity (2024)
• Networking Essentials — Cisco Networking Academy (2023)
• English Proficiency Certificate I & II — AUCA (2025)

LANGUAGES

- Kinyarwanda (Native)
- English (Professional)
- French (Basic)
- Kiswahili (Professional)

DATA COLLECTION LETTER

NDERA NEUROPSYCHIATRIC TEACHING HOSPITAL
BROTHERS OF CHARITY
P.O Box 423 Kigali - Rwanda; Tel: +250 788 827 364 / +250 781 447 928 / Toll free: 2575
Website: www.nderahospital.rw E - mail: ndera.hospital@moh.gov.rw

ETHICS COMMITTEE OF NDERA HOSPITAL

ETHICAL CLEARANCE

Ndera, March 28th, 2025
Ref: NNPTH/EC/2025/10

Student names: **Mr. Serge MUPENZI**

Institution: **AUCA** (Adventist University of Central Africa)

Your research project: **""AI-BASED CHATBOT FOR MENTAL HEALTH SUPPORT"** has been evaluated by the Ethics Committee of Ndera Neuropsychiatric Teaching Hospital.

Members of Ethics Committee of Ndera NPT Hospital			Involved in the decision		
N°	Names	Position	Yes	Absent	Withdrawn from the proceeding
1	Br. Eric Ferdinand TWIZEYIMANA	Director of Education, Research, CPD, and Quality Improvement, Co-chair	X		
2	Mrs. Gaudence MUKANDAYISENGA	Data Manager, Secretary of the Ethics Committee	X		
3	Mr. Lyevin BITANGA	Senior Systems Administrator, member	X		
4	Mrs. Jeanne d'Arc MUKAKALISA	Legal Affairs Officer, member		X	
5	Mrs. Eliane UMUBYEYI	Director of Human Resources and Administration Unit, member		X	
6	Mr. Jean de Dieu NSABIMANA	Public Relations & Communication Officer, member	X		
7	Mrs. Beatrice UWAMWEZI	Senior Nurse and In Charge of Accreditation	X		
8	Mrs. Scholastique UWINGABIRE	Representative of CARAES Butare		X	
9	Mrs. Clementine NIYOMWUNGERI	Representative of Icyizere Psychotherapeutic Center		X	

After reviewing your protocol and related documents presented during the Nder

Neuropsychiatric Teaching Hospital's Ethics Committee meeting of Thursday, March 28th, 2025 where quorum was met, **we hereby provide an approval for the above-mentioned protocol.**

Please note that the approval of protocol is valid for **12 months.**

You are responsible for fulfilling the following requirements:

1. Changes, amendments and addenda to the protocol or consent form must be submitted to the Ethics Committee for review and approval prior to activation of changes.

2. Only an approved consent form is to be used in the enrollment of participants.

3. All consent forms signed by subjects should be retained on file, the Ethics Committee may conduct audits of all study records, and consent documentation may be part of such audits.

4. A continuing review application must be submitted to the Ethics Committee in a timely fashion and before expiry of this approval.

5. Failure to submit a continuing review application will result in termination of the study.

6. Notify Ndera Neuropsychiatric teaching hospital's Ethics committee once the study is finished.

7. Submission of a **final copy of research findings** to the hospital is **mandatory.**

Date of Issue: **March 28th, 2025**

Expiration date: **March 27th, 2026**

Sincerely,

Br. Eric Ferdinand TWIZEYIMANA

Co-Chairperson of Ndera Neuropsychiatric Teaching Hospital's Ethics Committee

CC:

- Director General of Ndera Neuropsychiatric Teaching Hospital